

When infrastructure actually reaches people

Three peer-reviewed economics papers · two current LMIC studies and one Indonesia field-experiment classic. The following article pages are edited reading excerpts, not complete replacements for the originals.

LITERATURE PULSE

The shared result is that infrastructure does not diffuse value automatically. In the Philippines, the welfare return to mobile internet depends on whether network geography reaches households and supports local business activity, schooling, and flexible work. In interwar China, productivity gains from Japanese mills decay sharply with distance and depend on both foreign-firm quality and local absorptive capacity. In rural Indonesia, a sanitation technology that looks scalable on paper produces modest average take-up and narrow health gains because affordability, implementing agency, and social capital shape delivery. **Literature trend:** applied economists are increasingly measuring the last mile—geographic reach, organizational capacity, and implementation quality—rather than treating infrastructure provision as the treatment itself.

01 CURRENT / DIGITAL INFRASTRUCTURE AND HOUSEHOLD WEALTH · STARTS ON PAGE 2 · Journal of Development Economics · online 2026 · peer reviewed · CC BY

Mobile Internet Connectivity and Household Wealth in the Philippines

Does mobile connectivity translate into household wealth? The authors combine roughly 270,000 geocoded Philippine cell towers with the 2017 and 2022 Demographic and Health Surveys, instrumenting local tower density with distance to submarine-cable landing points. In the included public version, doubling tower density raises the standardized wealth index by about 0.04 standard deviations; point-of-interest activity, education, and women's flexible employment offer plausible mechanisms. Placebo, alternative-radius, and plausibly-exogenous checks are reassuring, but landing-point distance may affect wealth through other channels, DHS locations are displaced, and the local estimate excludes the densest urban areas where the instrument weakens.

02 CURRENT / FDI, INDUSTRIALIZATION, AND ECONOMIC HISTORY · STARTS ON PAGE 19 · Journal of Development Economics · online 2026 · peer reviewed · CC BY 4.0

FDI Spillovers in History: Interwar Japanese Investment in China's Cotton Spinning Industry

How local were the gains from early foreign direct investment? Using a 1925–1938 firm panel of Chinese cotton mills, geocoded Japanese Zaikabo plants, and a production function with firm and year fixed effects, the paper finds positive productivity spillovers concentrated within roughly 10–20 kilometers. Nearby Chinese firms also expanded spindle capacity, and gains were larger around older, larger, and more skill-intensive Japanese firms and where local absorptive capacity was stronger. The historical setting makes technology unusually comparable, but plant entry and location were not randomized, the estimation sample is small, and even careful within-Shanghai comparisons cannot eliminate time-varying local selection.

03 CLASSIC / PUBLIC HEALTH, IMPLEMENTATION, AND SOCIAL CAPITAL · STARTS ON PAGE 39 · Journal of Development Economics 138 · 2019 · peer reviewed classic · CC BY 4.0

Scaling Up Sanitation: Evidence from an RCT in Indonesia

Can a celebrated sanitation model survive national scale-up? Across 80 rural Indonesian villages, the authors randomize Community-Led Total Sanitation and measure toilets by inspection, parasites from fecal samples, anemia from blood, and child anthropometrics. The program modestly increased toilet construction and reduced roundworm intensity, but did not improve anemia, height, weight, or the overall health index. Gains were concentrated among nonpoor households and villages served by a better-performing resource agency, with social capital shaping take-up. The treatment randomization is strong; the implementing agency itself was not randomized, however, so agency-level heterogeneity is more suggestive than the average treatment effect.

Editorial note: references and appendices were omitted where possible; no included article contributes more than 15 text pages. Full citations and source links remain in the catalog. AI-written synopses were checked against the included source pages.

MOBILE INTERNET CONNECTIVITY AND HOUSEHOLD WEALTH IN THE PHILIPPINES

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ABSTRACT

Mobile internet has become a fundamental component of modern infrastructure. In this paper, we consider the impact of mobile internet connectivity on household wealth in the Philippines. We construct a granular measure of local mobile internet connectivity using comprehensive information on approximately 0.27 million geocoded cell towers, and identify causal impact through a novel instrumental variable based on proximity to submarine cable landing points. Our results suggest that mobile internet connectivity significantly increases household wealth, with effects that persist across education levels and are more pronounced in urban areas compared to rural ones. Combining individual survey datasets with Points-of-Interest data, we investigate mechanisms and demonstrate that improved connectivity stimulates activities in several key economic sectors that create employment opportunities. Additionally, mobile internet connectivity enhances individual educational outcomes and promotes female labor force participation, though predominantly in occasional or seasonal roles.

Keywords: mobile internet, cell tower, wealth inequality, Philippines

JEL codes: F14, J24, J63, L86, O33

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I Introduction

There has been a rapid increase in the availability and use of the internet in the developing world. As internet use has become increasingly common, the mobile phone has become the dominant mode of access, particularly among the poor. Internet use has had a profound economic impact, and studies in multiple contexts have shown important impacts (Aker and Mbiti, 2010; Forman et al., 2012; Akerman et al., 2015; Bahia et al., 2024). These benefits, however, are unevenly distributed across populations. Studies show that internet access generally has a greater positive impact on high-skilled employment and wages than lower-skilled labor markets (Hjort and Poulsen, 2019).

In this paper, we consider how the availability and quality of mobile internet access affects household wealth in the Philippines, a lower-middle-income country which saw a rapid rollout of mobile internet access in the 2010s, from initially low levels. Between 2010 and 2022 the share of the population using the internet went from 25 to 75 percent, and the number of mobile cellular subscriptions per 100 people went from 88 to 144.¹ The Philippines possesses a unique geography whereby it is comprised of over 7,000 islands. Given this, the cost and complexity of rolling out infrastructure has been an ongoing challenge. As such, fixed broadband internet in the Philippines has historically been expensive and slow. This has led to the vast majority of the population accessing the internet via their phones (Kanehira et al., 2024). Over this period, the Philippines experienced rapid economic growth, largely driven by its expanding service sector. This sector includes business process outsourcing, digital financial services, and e-commerce, all of which depend heavily on reliable internet access.

Since mobile internet coverage is not rolled out randomly, causally identifying its impact on household economic outcomes poses significant empirical challenges. In developing countries, for example, mobile internet usage is often poorly documented due to limited availability of high-quality survey data. Furthermore, failing to account for local socioeconomic conditions that affect both household wealth and cell tower deployment could result in omitted variable bias. Beyond that, another plausible endogeneity concern is reverse causality: while improved mobile internet access may boost household wealth, wealthier communities and metropolitan urban centers, by contrast,

¹ More details on the temporal trends of individual internet usage and mobile cellular subscriptions in the Philippines can be found in the following links: <https://data.worldbank.org/indicator/IT.NET.USER.ZS?locations=PH> and <https://data.worldbank.org/indicator/IT.CEL.SETS.P2?locations=PH>.

might also wield greater lobbying power to attract public investments in internet infrastructure or draw more commercial investments because of their larger market sizes.

To address these empirical challenges, our identification strategy begins by developing a granular measure of local mobile internet connectivity, proxied by the density of cell towers whose coverage scopes overlap with local communities. This measure is constructed using geospatial big data comprising approximately 0.27 million geocoded cell towers sourced from OpenCellID—a comprehensive, large-scale database providing precise locations and detailed information on cell towers. To overcome the endogeneity of mobile internet connectivity, we exploit the Philippines’ archipelago geography to develop a novel instrumental variable based on local communities’ geographical proximity to the nearest submarine cable landing points. The underlying intuition is that shorter distances to landing points reduce construction costs associated with expanding internet infrastructure, thereby influencing geographic patterns of mobile internet connectivity.² Indeed, we find evidence that better access to these landing points leads to substantial increases in mobile internet connectivity. Moreover, conditional on province fixed effects, the associations between distance to submarine cable landing points and various local socioeconomic factors—such as population density, nightlight luminosity, and livestock density—are statistically insignificant, supporting the plausibility of the exclusion restriction assumption.

Based on this approach, we first examine the overall impact of mobile internet connectivity on household wealth, using household survey data from the 2017 and 2022 waves of the Demographic and Health Survey (DHS). We find that better access to mobile internet is positively associated with household wealth, as measured by the DHS wealth index, which is constructed using principal component analysis of household asset ownership. Consistent with the relevance assumption of the instrumental variable approach, the first-stage results reveal a significant negative relationship between mobile internet connectivity and distance to the nearest

² Identification leveraging gradual rollout of submarine cables has become an important empirical technique. For instance, Hjort and Poulsen (2019) employ a Difference-in-Differences strategy based on proximity to terrestrial cables and the timing of gradual arrival of submarine cables to estimate the causal impact of high-speed internet on employment in Africa. We build on this literature with a focus on mobile internet access coming from proximity to cell towers, rather than fixed broadband, and introduce a new instrumental variable approach that uses distance to submarine cable landing points, rather than timing. While specific landing point locations are influenced by geography, distance of households from landing points is plausibly random and orthogonal to economic conditions, with controlling for province fixed effects, as we demonstrate in Section IV.

and explains how we measure the core variables that are used in our analysis. Additional data sources are introduced later when they are used for the first time.⁶

III.A Cell Towers and Mobile Internet Density

We source mobile internet data from OpenCelliD, a large-scale global open database providing extensive information on cell towers and their locations.⁷ The OpenCelliD database records information for each cell tower, including the generation of broadband cellular network technology (radio types: GSM/2G, UMTS/3G, LTE/4G, and NR/5G), the country and region where the cell tower is located, and its geographic coordinates (longitude and latitude). The database also flags whether the geographic coordinates of cell towers are provided directly by telecom companies or derived from user-submitted data, which combines the signal strength received by user's mobile equipment with its positional information.⁸ Additionally, the database records the date each cell tower was first added into the database and when it was seen. We restrict our analysis to cell towers located in the Philippines that were first added into the database between 2008 and 2022.⁹ As a consequence, we ultimately obtain 265,246 georeferenced cell towers, all with geographic coordinates derived from user-submitted data, and containing three radio types – GSM, UMTS, and LTE. We use the date each cell tower was first added to the database as a proxy for its construction time and we assume no cell towers are decommissioned due to a lack of such information. While we acknowledge the data limitations regarding the locations, construction times, and active durations of cell towers, OpenCelliD, to the best of our knowledge, offers the most accurate and freely available data on cell tower locations in the Philippine context. Nonetheless, as discussed in the next section, our empirical strategy is well-equipped to account for these potential measurement errors.

Using the cell tower data, Figure 1 illustrates the cumulative number of cell towers in the Philippines from 2008 to 2022, categorized by radio types. We observe a rapid roll-out of cell

⁶ Details on auxiliary data sources are provided in Appendix A. The Appendix is available at <http://dx.doi.org/10.22617/WPS250440-2>.

⁷ For more details and information on the methodology, visit <https://opencellid.org/>. The OpenCelliD project is licensed under a Creative Commons Attribution-ShareAlike 4.0 International License.

⁸ If the geographic coordinates of a cell tower are obtained in the second way, the database also includes the number of (user-submitted) samples or measurements processed to determine the location, as well as a radius indicating the range within which the actual location is likely to fall.

⁹ We found only one cell tower in the Philippines that was added to the database before 2008, which is likely to come out as an erroneous entry.

III.B Submarine Cable Landing Points

Our identification strategy relies on proximity to submarine cable landing points across the Philippines as an instrumental variable. Landing points are coastal sites where submarine internet cables connect to terrestrial networks. These cables enable the transmission of large volumes of data across oceans, linking countries to the global internet infrastructure. At the landing points, data is transferred to land-based systems such as fiber-optic networks, data centers, and mobile networks. We collect data of georeferenced submarine cable landing points from Infrapedia, an open-source database to provide complete and versatile infrastructure map of the Internet.¹⁵ We obtain the geographic coordinates (longitude and latitude) of submarine cable landing points and detailed information about the submarine cables connected to them, including the years these cables became operational. For each landing point, we assign its ready-for-service time based on the earliest operational year among the connected submarine cables. Our primary instrumental variable is the Euclidean distance from a DHS cluster's centroid to the nearest existing submarine cable landing point.¹⁶

Figure 3 depicts the spatial distribution of submarine cable landing points across the Philippines, color-coded by their ready-for-service years.¹⁷ We see that the majority of submarine cable landing points were constructed either before 2003 or between 2017 and 2022. We focus on the most recent set of submarine cable landing points, which are expected to provide a stronger instrumental variable (as explained in the robustness check section). To test whether these landing points were placed in wealthier areas, we examine provincial GDP data from 2018 to 2022. Specifically, we analyze whether provinces with higher GDP were more likely to host new landing points during this period. To examine this, we calculate the number of landing points constructed in each province and run a Poisson regression with province and year fixed effects, using robust standard errors. The p -value for the coefficient on provincial GDP is 0.763, suggesting that landing points were not disproportionately located in wealthier provinces. Rather, their locations are more likely determined by proximity to international or national

¹⁵More details are provided in the link: <https://www.infrapedia.com>.

¹⁶To minimize distortion in distance measurements, we perform geocomputation based on the WGS84 UTM Zone 51N projection system.

¹⁷Appendix Table F.1 lists the specific location name and ready-for-service year for each landing point. Appendix Figure F.1 provides a snapshot of the submarine cable network across the Philippines, taken from the Infrapedia database.

analysis. Summary statistics for these variables, along with our primary variables, are reported in Appendix Table G.1.

IV Empirical Strategy: Instrumental Variable from Submarine Cables

Our parameter of interest is the medium- to long-term impact on household wealth accumulation, stemming from the staggered rollout of cell towers and the resulting variation in mobile internet exposure across localities in the Philippines. Specifically, we examine whether mobile internet connectivity contributes to improving households' wealth status by leveraging exogenous variations in cell tower density driven by the extent of remoteness from the submarine and territorial cable network. In other words, we instrument mobile internet density using the Euclidean distance from the centroid of each DHS cluster to the nearest existing submarine cable landing point. The Philippines is an archipelagic country that depends on a network of submarine and land-based cables to provide internet access. Our instrumental variable is based on the idea that areas farther from cable landing points face higher costs for building internet infrastructure. As a result, these areas tend to have fewer cell towers. We assume that, after conditional on a key set of covariates and focusing on comparisons within a small geographic area, the distance to the nearest landing point is not correlated with other factors that affect household wealth.

Before examining the identification assumption in detail, we first describe our baseline econometric model, which is estimated using two-stage least squares (2SLS):

$$Wealth_{icpt} = \mu_t \alpha_p \gamma_0 \cdot \widehat{Mobile\ internet\ density}_{cpt} X'_{icpt} \Omega_0 \epsilon_{icpt}, \quad (1)$$

$$Mobile\ internet\ density_{cpt} = \mu_t \alpha_p \gamma_1 \cdot Distance_{cpt} X'_{icpt} \Omega_1 \epsilon_{cpt}, \quad (2)$$

where $Wealth_{icpt}$ denotes household wealth status, measured in quintiles of the DHS Household Wealth Index on a scale from 1 (poorest) to 5 (richest), for household i , residing in DHS cluster c , within province p , and interviewed in wave t (2017 or 2022).¹⁹ We standardize the quintile dependent variable for ease of interpretation. Our primary explanatory variable is mobile

¹⁹As mentioned above, the two waves of the DHS survey correspond to periods following a surge in the number of cell tower rollouts across the Philippines, at least 9 years after the construction of cell towers began in the country (see Figure 1 for details). This timing allows us to study the medium- to long-term impact of mobile internet connectivity on household wealth accumulation.

internet density, *Mobile internet density*_{cpt}, defined as the log of cell tower counts per 1,000 residents for each DHS cluster. γ_0 therefore denotes the parameter of our interest. The instrument, *Distance*_{cpt}, denotes the Euclidean distance from the centroid of each DHS cluster to the nearest existing submarine cable landing point.²⁰

Our specifications also include fixed effects for the survey wave and province (μ_t and α_p) to capture overall differences in household wealth across the time and regional dimensions. For example, survey wave fixed effects enable comparisons within each wave, thus accounting for the issue that the Wealth Index constructed from a mix of household assets might be statistically inconsistent between waves due to changes in the composition of assets involved. Moreover, the importance of distance in influencing cell tower rollouts may diminish over a broad geographic scale (e.g., mobile internet operators might prioritize distant but economically or politically significant areas despite higher construction costs). By incorporating provincial fixed effects, we narrow the focus to comparisons among DHS clusters within a relatively small geographic scale, where distance is more likely to play a crucial role as a determinant of cell tower construction. Additionally, as we demonstrate below, focusing on a smaller geographic scale increases the likelihood that DHS clusters are balanced across other socio-economic factors that might also affect cell tower deployment.

We control for a rich set of covariates at both the cluster and household levels, denoted as X'_{icpt} . Our cluster-level controls include: (i) a dummy variable indicating whether DHS clusters are situated in urban areas; (ii) population density and nightlight luminosity, which broadly capture local economic development (urbanization and economic activities); (iii) rainfall and temperature, reflecting overall climatic conditions that may influence both economic activities and the feasibility of cell tower construction (e.g., lightning strike intensity has been shown to impact mobile phone coverage (Manacorda and Tesei, 2020)); and (iv) slope of terrain that could influence the strength and quality of mobile internet signals (e.g., Wang, 2021). At the household level, we control for the number of household members, as well as the age, gender, and educational attainment of the household head, as these factors are likely to directly affect household wealth status.

²⁰For the 2017 DHS survey wave, we calculate the distance for DHS clusters based on landing points that were operational in 2017 (i.e., those constructed before 2017), taking out of consideration those that became operational only after that year. For the 2022 DHS survey wave, we include all landing points that were operational by 2022. Our results, however, remain robust when using alternative sets of instruments, such as the distance to landing points established before 2003 (see columns (1) and (2) of Table I.1 for more details).

Instrument Relevance To examine the validity of our instrumental variable, we begin by testing whether remoteness from submarine cable landing points reduces mobile internet density. Figure 5 presents binned scatterplots illustrating the relationship between the local density of cell towers and the distance to the nearest submarine cable landing points across DHS clusters. In addition to considering all cell towers collectively, we further disaggregate them by radio types—GSM, UMTS, and LTE—and calculate the corresponding measures of mobile internet density for each type. The descriptive results align with our theoretical expectations, showing a negative association between the instrument and mobile internet density, regardless of the cell tower type used for measuring density. In our subsequent estimations, we provide 2SLS regression results including first-stage estimates. Together, these results indicate sufficient instrument relevance.

Exclusion Restriction A major identifying assumption of our empirical approach is that the distance to the nearest existing submarine cable landing point influences household wealth solely through its effect on mobile internet density, conditional on a key set of covariates and fixed effects. In the previous sections, we demonstrated that landing points are not preferentially located in wealthier provinces but are instead determined by geographical factors and the need to integrate with the global submarine cable network or internal internet infrastructure. However, this finding does not fully establish the exogeneity of distance to landing points, as this measure may also capture proximity to coastlines, which is therefore closely correlated with local economic development and violate the exclusion restriction assumption. Table 1 investigates the associations between the distance to submarine cable landing points and various local socio-economic factors across DHS clusters. In columns (1) and (2), we examine the relationships with population density and nightlight luminosity using the full set of DHS clusters, whereas columns (3) through (7) focus on livestock density, restricting our focus to rural clusters. Indeed, we find that in the absence of province fixed effects, DHS clusters located farther away from submarine cable landing points tend to exhibit lower population density, dimmer nightlight luminosity, and lower density of livestock such as pigs and chickens. However, these differences across socio-economic dimensions diminish once province fixed effects are included, suggesting that socio-economic factors are more likely to be balanced with respect to distance from submarine cable landing points when comparisons are made within a relatively small geographic scale. Consequently, we include province fixed effects in all subsequent 2SLS estimations.

In our empirical exercises as described below, we perform a range of robustness checks and placebo tests to further evaluate the validity of the exclusion restriction. First, we utilize the DHS survey conducted in 2003 – prior to the rollout of cell towers in the Philippines but after the establishment of an early wave of submarine cable landing points – to assess whether the distance to these landing points predicts household wealth at that time. If distance is found to influence household wealth in 2003, it is plausible that it operates through channels other than mobile internet density, therefore invalidating the exclusion restriction. Additionally, we perform placebo tests by randomly assigning the baseline instrument values to other DHS clusters within the same survey wave or within the same province. A valid instrument should reveal that these placebo instruments exhibit weak instrument characteristics and lack significant association with household wealth status. We also employ the plausibly exogenous framework proposed by Conley et al. (2012), allowing our instrumental variable to exert direct effects on the main outcomes of interest. This method enables us to assess the sensitivity of our findings to varying degrees of instrument invalidity. We provide detailed discussions of these robustness checks and the associated results in the following sections.

V Empirical Results

V.A Impacts on Household Wealth

Our empirical analysis begins by estimating our baseline specification, as defined in Equation (1) and Equation (2), leveraging the distance to the nearest existing submarine cable landing point as an instrument for mobile internet density, while controlling for survey wave fixed effects and province fixed effects. As a benchmark, we report simple OLS results to illustrate the endogenous correlational relationship between mobile internet density and the household wealth index constructed based on asset ownership. Both the OLS and 2SLS results are presented in Table 2.

We find a positive and statistically significant association between mobile internet density and the standardized household wealth index, a finding that holds across all specifications as we sequentially introduce controls for locality (column (1)) and household characteristics (column (2)), and mobile phone ownership which we include to address concerns that our results may be

skewed by access to mobile devices, rather than access to mobile internet infrastructure like cell towers (column (3)). While the coefficient estimates for mobile internet density reduce slightly in magnitude, they all remain positive and significant at the 1 percent level.

Next, we present the instrumental variable estimates in columns (4) to (6). Panel A shows that the second-stage results suggest a positive causal effect of mobile internet density on household wealth. This finding is consistent across all three models, which control for locality-level and household-level confounders, including mobile phone ownership. Consistent with our theoretical expectations, we can see a significantly negative relationship from the first-stage results in Panel B: as the distance from a submarine cable landing point increases, mobile internet density declines. Furthermore, the first-stage Kleibergen-Paap Wald rk F statistics all exceed the standard threshold of 10, and the Anderson-Rubin tests reject the null hypothesis. Together, these results indicate that our instrument is a strong predictor of local cell tower density.

One can also see that our 2SLS estimates are notably larger than the OLS estimates (approximately tenfold). In the fully specified model, doubling the number of cell towers per 1,000 people within neighborhoods is associated with a 0.04 ($\approx 0.145 \times \log 2$) standard deviation increase in the household wealth index. This implies that the OLS approach underestimates the role of mobile internet density in improving household wealth status. Beyond the influence of omitted variables, measurement errors (and thus attenuation bias), and reverse causality, as discussed earlier, the larger 2SLS estimates may reflect local average treatment effects (LATE) specific to areas where the construction of cell towers is primarily determined by ease of access to the cable network, a point we will elaborate on in the next section.

A natural concern regarding our measurement of mobile internet density is whether coverage, based on prescribed buffers around cell towers (i.e., proximity of users to mobile towers) and the average count of locally built cell towers, effectively captures mobile internet connectivity or the quality of internet service available. We examine this issue through two approaches. First, we acknowledge that distance-based measures of mobile internet access may not capture connectivity consistently across different localities due to factors such as geographical topography, climate, and other local conditions. For instance, flat areas farther from cell towers than mountainous regions may still experience better mobile internet access (it is important to note that our inclusion of terrain slope as a control variable could partially account

quality. Taken together, the results indicate that the density of mobile internet cell towers measured in our analysis positively impacts household wealth conditions, with the effect primarily driven by mobile internet connectivity and service quality.

Robustness Checks We now turn to evaluating the robustness of our empirical strategy, probing the validity of distance to submarine cable landing points as an instrument for mobile internet density. As demonstrated below, our findings are robust to a range of placebo tests and to variations in several salient dimensions of our measurements and estimation approaches.

First, we provide plausible empirical evidence on exclusion restriction that mobile internet access is the only channel through which distance to submarine cable landing point impacts household wealth, conditional on a crucial set of control variables. To this end, we conduct reduced-form estimations by regressing the standardized household wealth index on distance to submarine cable landing point, using DHS data from the combined 2017 and 2022 waves and more importantly, using DHS data from the 2003 wave, a period when submarine cables existed but mobile cell towers had not yet been rolled out in the Philippines.

Appendix Table H.1 presents the results, with columns (1) and (2) showing the reduced-form relationship between distance to submarine cable landing point and household wealth in 2003 as a placebo test (note that mobile phone ownership is not controlled for, as this variable was unavailable in that wave), and columns (3)–(5) reporting the relationship for the 2017 and 2022 waves. Conditional on province fixed effects, and local and household characteristics, one can see the expected negative and statistically significant relationship between household wealth and distance from submarine cable landing point in the post-rollout period; however, while there is a negative association before the rollout, it was not statistically significant at the conventional level. The absence of a significant relationship between household wealth and distance from submarine cable landing point before the rollout of mobile cell towers, but its emergence after the rollout, suggests that mobile internet access is likely the sole channel through which the distance influences household wealth.

Second, we further examine the assumption of instrument exogeneity by following the plausibly exogenous framework proposed by Conley et al. (2012). The main idea of this approach is to allow our instrumental variable to have direct effects on the main outcomes of interest; specifically, the instrumental variable is involved in the second-stage regression with a coefficient δ . If the exclusion

technology usage behavior (e.g., using internet for e-commerce vs. addiction to video games).²⁵ The Philippines, an archipelago with significant regional and geographical disparities, offers a case study for examining how internet access can influence wealth distribution.

We begin our analysis of digital inequality by examining the impacts of mobile internet connectivity across urban and rural areas. To capture potential differential effects on household wealth, we split our sample into urban and rural households, allowing mobile internet connectivity to have distinct impacts depending on the area of residence. Comparing columns (1) and (2) in Table 4, we observe significantly positive effects of mobile internet density on the standardized household wealth index in rural clusters, while the effects in urban areas are positive, with a coefficient estimate approximately 2.6 times larger in magnitude than that for rural areas, but statistically insignificant. Columns (3) and (4) focus on urban clusters using an alternative definition of urbanization. Column (3) restricts the sample to households located in Barangays (local administrative units at the third level in the Philippines) classified as cities by the Global Human Settlement Layer (GHSL) project. Column (4) expands the sample to include households in Barangays classified as dense towns by the GHSL project. The results indicate that changing the urban definitions does not yield significant effects for urban households. Indeed, the Kleibergen-Paap Wald rk F statistics and AR tests suggest that the 2SLS estimates for urban households might suffer from a weak instrument.

It is important to note that the weak instrument issue for urban clusters is expected, as telecom companies often prioritize densely populated, politically or economically important cities, regardless of their distances from submarine cable landing points. To further assess where our previously estimated LATEs apply, we focus on areas beyond the rural regions where we have found robust estimates with a strong instrument. Continuing with Table 4, our analysis proceeds by focusing on households located in urban clusters while excluding those in Barangays with large population sizes, using data from the GHSL project. Specifically, we progressively exclude

²⁵The impact of internet access on inequality has been the subject of extensive and nuanced debate. For example, one perspective suggests that the internet and modern Information and Communication Technologies (ICTs) have the potential to reduce inequality by spreading economic activities and expanding job opportunities across geographic boundaries (Friedman, 2007). This can lead to a more equitable distribution of employment opportunities. However, an opposing viewpoint argues that the advent of the internet has led to “skill-biased technological change”, favoring individuals with higher levels of education and skills (Akerman et al., 2015), therefore, resulting in an increase in income inequality. Empirical evidence on the effects of internet access on inequality remains limited, particularly in the context of developing countries.

DHS cluster from a pool of the 2017 and 2022 waves, using POI density, defined as the number of POIs per 1,000 people.²⁷

With POI density as the outcome variable, we conduct our IV estimations, controlling for wave fixed effects, province fixed effects, and cluster characteristics as in our household-level analysis. We present the estimates in Table 5, with different columns focusing on different types of POIs. To mitigate weak instrument issues in large urban areas as discussed above, Panel A excludes clusters located in Barangays with population sizes exceeding the 99th percentile, while Panel B further excludes urban clusters located in Barangays with population sizes exceeding the 80th percentile. From results in both panels, one can see that the density of POIs in the sectors of Business and Professional Services, Dining and Drinking, Retail, and Travel and Transportation differentially increase in localities with higher mobile internet density. However, we find no evidence of a positive impact on the sector of Arts and Entertainment. This suggests that mobile internet access indeed boosts crucial economic activities that could provide more job opportunities and, in turn, improve household wealth.

Subsequently, we analyze the supply side of the labor market, evaluating the impact of mobile internet connectivity on individual employment status. We examine employment outcomes using the “Individual Record” (IR) datasets of DHS survey, which primarily focus on women in households (we restrict the sample to women aged 18 and above). We first estimate our baseline 2SLS model for female respondents, using binary indicators as dependent variables to capture different employment statuses: whether the female is employed during the seven days preceding the survey interview or at any point in the past 12 months. As shown in columns (1) and (2) of Table 6, we find that females are more likely to be employed—either recently or during the 12 months prior to the interview—in areas with higher levels of mobile internet connectivity. Columns (3) to (5) focus on employed women and examine how mobile internet connectivity influences their mode of employment, using binary dependent variables indicating whether the respondent was employed year-round, seasonally, or occasionally. Interestingly, the results show a negative association between mobile internet connectivity and year-round employment among women, while seasonal

²⁷For DHS clusters in 2017, POIs considered were those with entry dates before 2017 and not marked as closed in the database, while for clusters in 2022, POIs were those recorded before 2022 and had not been closed by then. It is important to note that the date a POI entered the database does not necessarily reflect its actual opening date, just as the recorded close date may not precisely indicate when the POI ceased operations. While the ideal approach would be to include only active POIs, data limitations prevent us from doing so.

and occasional employment are positively associated. This pattern suggests that mobile internet may enable more flexible work arrangements for female workers.

Given that only 50 percent of women are currently employed in the Philippines, these findings underscore the need for targeted interventions—such as digital literacy programs, childcare support, and skill-building initiatives—to help women fully capitalize on the economic opportunities enabled by internet access. The Philippine DHS survey does not include “MR” files focused on male respondents, limiting our ability to analyze male employment outcomes. A promising avenue for future research is to systematically examine whether the wealth-enhancing effects of mobile internet access also operate through its impact on male employment, and to assess gender disparities in the economic benefits of digital connectivity. Men may be better positioned to capitalize on internet-enabled economic opportunities, possibly due to existing gender disparities in the labor market, digital skills, or sectoral employment patterns.

We now consider educational outcomes, utilizing the “Personal Record” (PR) datasets, which provide individual-level information on household members (the sample is restricted to individuals aged 18 and above). The 2SLS estimates are reported in columns (6) and (7) of Table 6, respectively. Our outcome variables of interest are a binary indicator for attaining at least secondary education (column (6)), and the number of years of educational attainment (column (7)). The results indicate that higher mobile internet density significantly increases the probability of attaining secondary education and the total years of schooling, suggesting that improved internet access could perhaps facilitate better educational outcomes by providing access to online learning resources, educational materials, and information on schooling opportunities. This implies that mobile internet may play a role in reducing long-term human capital inequalities, especially in rural or underserved areas.

VI Conclusion

The growing use and importance of internet access has had profound economic impacts across the world. This paper has considered the impact of growing mobile internet access on household wealth in the Philippines. We use the staggered rollout of cell towers and an instrumental variable based on distance to the nearest submarine cable landing point for causal identification. Our results show that mobile internet connectivity leads to higher household wealth. Our estimates

represent local average treatment effects that exclude the most densely populated urban areas where the instrument is relatively weak in strength. Within this sample, all groups appear to benefit from mobile internet connectivity, with more pronounced effects observed in urban areas compared to rural ones. We also find positive effects across varying levels of educational attainment, with the magnitude of the effects increasing as education levels rise, although the estimates are less statistically significant for higher education groups. Overall, our findings show that mobile internet helps raise household wealth and that the gains are broadly shared.

We test mechanisms that these benefits of mobile internet connectivity likely operate through its role as a fundamental component of modern infrastructure, and find that it stimulates economic activities in crucial sectors which could generate employment opportunities. On the other hand, our results indicate that mobile internet connectivity increases female labor force participation in occasional and seasonal employment, while reducing engagement in year-round jobs. It also appears to enhance human capital accumulation through improved access to information and digital learning tools. Together, our findings underscore the multifaceted value of digital connectivity in promoting inclusive economic development, and contribute to ongoing debates about the distributional impact of mobile internet access (e.g., Hjort and Tian, 2025). They highlight that there are important payoffs from investments in digital infrastructure but that, while important, these investments are not enough. Complementary investments such as digital skills training and improved access to mobile devices will help ensure benefits are more widely and equitably shared.

The Philippines provides a useful setting for our identification strategy; however, caution is warranted in generalizing the results beyond this context. As an archipelagic country, internet access may play a more important role in connectivity and economic activity compared to more geographically contiguous countries, which makes it well-suited for our approach. Moreover, the Philippines' service-oriented economy and its considerable reliance on remittances may shape the relationship between mobile internet access and household wealth in ways that differ from other settings. In addition, our findings reflect the rollout of a specific technology during a specific time period, and these effects may not persist as technologies and usage patterns evolve. Nonetheless, our results provide strong evidence that improved and more widespread mobile internet access can contribute to increased household wealth. And a promising direction for future research would

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**FDI Spillovers in History: Interwar Japanese
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DISCUSSION PAPERS

FDI Spillovers in History:

Interwar Japanese investment in the Chinese cotton industry

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Abstract

In this paper we use comprehensive historic firm level data for 1925 to 1938 to estimate productivity spillovers from Japanese textile companies' affiliates in China (Zaikabo) to local cotton producers in China. We geo-localized firms in order to capture the important role of distance in facilitating productivity spillovers. Our results provide clear evidence for positive productivity spillovers from Zaikabo to local Chinese firms. This goes hand-in-hand with a change in production technology towards greater use of capital (spindles). We also find that spillovers are very localised, being strongest within a radius of up to 10km around the Zaikabo. Furthermore, evidence for spillovers is particularly strong for firms in Shanghai. Our paper is the first to provide evidence for such spillovers from foreign firms in a historical context.

JEL Codes: F23, N65

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1 Introduction

Attracting foreign direct investment is frequently seen as an important means to foster overall economic development, or the growth of particular industries, in host countries. This is reflected in policies trying to boost inward investment (UNCTAD, 2018). Despite extensive research in the last decades on potential positive spillover effects from multinationals,¹ the jury is still out on whether such effects are there, and what the exact mechanisms are.

A convincing search for such effects is made difficult by the complexity of modern-day industry. Different industries use different technologies, and organise their production chains in different ways. Looking across various industries (which most empirical studies of spillovers do) therefore makes it difficult to tease out what the exact mechanisms are through which spillovers from FDI may happen. Also, it is challenging to disentangle possible positive knowledge transfers from negative competition effects crowding out local firms. In addition, the search for spillovers is further muddled by selection effects where it is important to clearly distinguish firms' locational choice from spillovers – do multinationals simply locate in industry-region clusters with favourable characteristics, or do they actually benefit local firms? Hence, to identify such effects more convincingly, it may make sense to look at a simpler world in history. This is what this paper sets out to do: We use comprehensive and unique historic firm level data for 1925 to 1938 to estimate productivity spillovers from Japanese textile companies' affiliates in China (Zaikabo) to local cotton producers in China.

We follow the well-established literature in modern contexts that tries to empirically identify “productivity spillovers” from foreign to domestic firms (Lu et al., 2017; Girma et al., 2015 look specifically at China). In the setting here we look at spillovers from the more advanced Japanese firms to local Chinese establishments. To this end we geo-localized and digitized information on all Japanese textile companies' affiliates in China (including Manchuria) as well as other local firms in China during the period 1925 until 1938. Importantly, apart from output and employment, the data contains detailed information on the quantity and type of capital stock employed, allowing us to accurately measure the nature of technology utilized by the Japanese foreign affiliates.

¹ Recent examples are Amiti et al. (2024), Jiang et al. (2024), Abebe et al. (2022), and Fons-Rosen et al. (2018). Görg and Strobl (2001) and Keller (2021) provide extensive surveys.

Being able to pinpoint the exact location of both the Japanese headquarters and plants and their Chinese counterparts importantly allows us to capture the role of distance in facilitating any productivity spillovers. This represents a novelty of our study, as the role of distance is generally neglected even in modern spillover studies on FDI. Rather, the focus of the spillover literature has been primarily on spillovers within an industry, or across vertically connected industries, with a few studies also considering spillovers within geographic units – usually defined as fairly large administrative region (e.g., Girma et al. 2015; Aitken and Harrison, 1999). Our fine-grained data with geolocalized information on both locations of Japanese FDI firms as well as local Chinese firms allows us to dig much deeper into the role of distance, using distance-weighted spillover variables as are commonly employed in the literature on knowledge spillovers (e.g. Jaffe and Trajtenberg, 1996; Figueriredo et al., 2015).

Some other papers investigated Japanese FDI in China in terms of location but without directly investigating spillovers, as, e.g., Belderbos and Carree (2002) and Kim et al. (2010)² However, none of these studies directly consider the role of distance between local and multinational firms in determining productivity spillovers, despite the crucial importance of distance in facilitating technological spillovers being well established in the literature (Bloom, et. al, 2013). Arguably, and in line with our results, geographic proximity of foreign and domestic firms is likely to have been even more important in terms of facilitating spillovers in historical settings with less developed transport and communication infrastructure.

Another attraction of our setting is that the time period of analysis (1925 to 1938) covers a turbulent time in the Japanese-Sino relationship. More precisely, while anti-Japanese movements (e.g., strikes in Japanese multinationals, non-use and non-sales campaigns against Japanese products) had already been happening since the 1910s, the “Mukden Incident (Manchuria Incident)” in 1931 resulted in Japanese troops occupying part of Chinese territory and the foundation of a nation, Manchuria, in 1932 under Japanese political and military control.³ As a response, large riots occurred in Shanghai and Japanese military troops were

² Belderbos and Carree (2002) investigated the location of the Japanese affiliates in China in terms of firm heterogeneity and found that small size firms tend to locate closer to Japan but have less benefits from local Chinese market. Kim et al. (2010) investigated the exit of the Japanese subsidiaries in China from 1979 to 2001 and found that exit rates are affected by parent firms’ operation experience in the industries and geographical distance from other Japanese subsidiaries.

³ The Mukden Incident is a Japanese military action, an explosion that destroyed South Manchuria Railways on September 18th 1931. The Japanese military accused Chinese dissidents of the act and used it as a pretext for the military invasion of North East China.

effect of foreign direct investment (FDI) from potentially endogenous location choices by Japanese firms.

As a possible mechanism of how firms boost productivity, we also observe an increase in capital stock, as reflected in the number of spindles employed by Chinese cotton firms. Although we cannot directly observe more about underlying mechanisms, our findings—particularly the highly localized nature of the effects—suggest that job-hopping within a shared labor market between Japanese and local Chinese firms may have played a role in facilitating knowledge transfer (Ashizawa, 2014, 2011). Regarding the Japanese firms themselves, we find that larger firms, both in terms of local plant size and total size as multi-plant enterprises, generate stronger spillovers. Additionally, older firms and those with a more international orientation—evidenced by headquarters located outside Japan—tend to have a greater spillover effect on local Chinese firms.

The remainder of the paper is organized as follows. We describe the relevant background of the cotton industry, including the role of Japanese firms, in China for our setting. The data used in our analysis is describe in Section III. Our methodological approach is outlined in Section IV. Results are discussed in Section V. In the final section we provide some concluding remarks.

2 Historical Background

2.1 The Development of the Japanese Cotton Industry Post-Autarky

Japan had been an autarky for around 250 years in the Edo period. It finally opened several ports to trade in 1858 as a result of the US–Japan Treaty of Amity and Commerce. Consequently, silk and textiles quickly became major export products (see Yamazawa and Yamamoto, 1974; Okubo, 2007).⁸ Trade continued to increase, with the process accelerating during the Meiji Restoration (1870s-1880s), where the core industries were silk reeling, and weaving, as well as cotton spinning and fabric manufacturing. In terms of cotton spinning, after Osaka Boushiki Co. was founded in 1882, several large-scale companies followed and successfully produced high-quality cotton textile products by introducing advanced foreign technology and machines (Nakamura, 1971). By the early part of the 20th century, the textile

⁸ The opening of trade also drastically changed the domestic market; see Bernhofen and Brown (2005) for details.

2.4 Background on productivity spillovers from Zaikabo

According to theories of the multinational enterprise (such as Markusen 1995), multinationals have a technology advantage vis-à-vis domestic firms, which can then “spill over” to local firms (e.g., Görg and Strobl, 2001). Hence, an important condition for productivity spillovers from Zakaibo to Chinese local firms (“Minzokubo”) to occur is that the former had substantial productivity advantages over the latter. In this regard, as shown in Takamura (1982: p181), while Zaikabo's productivity was lower compared to Japan (48-52% of Japanese domestic production), it still outperformed local Chinese firms, which operated at only 55-70% of Zaikabo's productivity levels. As pointed out by Takamura (1982: pp.181–184), there are a number of reasons for this. Firstly, Zaikabo benefited from using modern and more advanced machinery, in particular new technologies like high draft machines, while Chinese local firms (Minzokubo) used outdated, often by decades, machines.¹⁴ Second, Zaikabo had more efficient management systems. They implemented a direct control employment system, akin to the standard Japanese management system, which ensured better management and coordination of employees, often involving a piece rate wage system tied to class-based wages.¹⁵ In contrast, Chinese local firms relied on the “Houshin Sei” employment system, which involved contractors hiring workers from rural areas, leading to inefficiencies, poor machine management, and an overall lack of effective supervision. Third, Japanese networks provided Zaikabo with access to lower-cost capital from Japanese banks and cotton at reasonable prices from Japanese trading companies, whereas Minzokubo firms often had limited access to capital and had to rely on higher-cost material inputs. Fourth, Minzokubo faced greater taxation than Zaikabo. For instance, in the Shanghai International Settlement, Zaikabo firms were levied only land tax and small contributions to the Japanese Residents Union, but otherwise no taxation. Thus, they enjoyed more favorable taxation conditions than the Minzokubo due to their extraterritoriality.

Given that Zaikabo had a productivity advantage compared to local firms, one may expect the potential for spillovers to be present. Indeed, there is evidence that Zaikabo aggressively transferred technologies and know-how from headquarters and main plants in Japan

¹⁴ In Minzokubo, extremely old spinning machines, used for 26 to 47 years, accounted for 30%, while those used for 16 to 25 years made up 14%. While over 90% of Zaikabos used machines with electric motors, among the 45 Chinese factories, 29 used electric motors, 12 used steam engines, and 4 used both. (Takamura, 1982: p.182)

¹⁵ One should note that while this was efficient, it did occasionally lead to racial tensions.

3 Data sources

In order to examine more comprehensively whether Chinese producers benefited from the presence of Zaikabo in their geographic vicinity, we have collected unique data at the firm level relating to the activities of Zaikabo and Chinese local cotton producers (Minzokubo). The data come from the *Handbook of Textile Industries* (“Boushoku Yoran”) which is published annually by Boshoku Zasshi-sha, and *Statistics on Japanese Spinning Industry*, (“Bouseki Jijo Sankosho”) published by Dai-Nihon Bouseki Rengokai (Japan Association of Cotton Spinning). We have assembled the data for the period 1925 to 1938. Both data cover all Japanese textile companies and their plants in Japan, their affiliates in China (including Manchuria) as well as other local firms in China.

We are able to obtain information on firms’ addresses (headquarters and plants), their total output in terms of cotton value, the number of spindles and looms, capital assets, number of employees as well as the year of foundation of the firm. This is available for Zaikabo and Chinese firms. Before 1935 the dataset covers information in Tan and afterwards in Yards, hence we adjust the data accordingly and measure everything in Yards. Likewise, capital assets for Chinese firms are in Chinese currency (silver) units (Tael and Yuan). We converted tael to yuan.

In order to geo-locate firms, using information on the firms’ addresses we gather detailed information on longitude and latitude. We determine the coordinates of plant location in Shanghai from historical maps on the Shanghai Concession, “Plan of Shanghai” published by the War Office, Geographical Section (1937), “Saishin Shanghai Chizu (New Shanghai Map)” published by Osaka Asahi Newspaper (1932). These maps include all street names as well as locations of companies and plants, hotels, banks, apartments, shops, transportation infrastructure and public places (such as hospitals, schools, temples, and churches). We can also identify the location of the Japanese residential place, “Little Tokyo”. Since most streets are the same as current ones, we use GIS to measure coordinates. We can thus identify the exact location of each plant in Shanghai.

To do the same for firms located outside of Shanghai, we use the information on firm location from the firm data, which is available at the town level. We then use the name of the town and determine the centre of the town by GIS. Figure 1, Panel A illustrates the locations of Zaikabo in green and the location of Chinese local firms (Minzokubo) in red. This shows a

concentration of firms in the Eastern part and, in particular, in Shanghai. Panel B zooms in on Shanghai.

Figure 1: Locations of Zaikabo and Chinese Local Firms

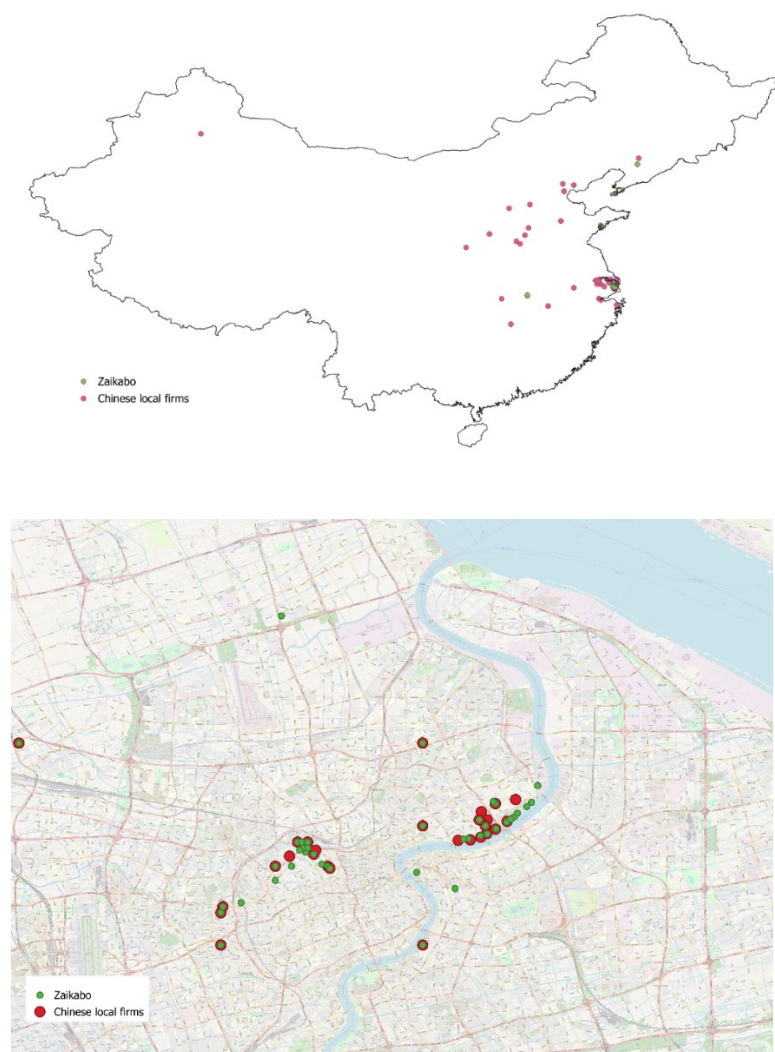


Table 1 shows descriptive statistics for Chinese local firms (Minzokubo) and Zaikabo in our data. There is a substantial variation in firm size as Chinese local firms employ between 270 and 5,923-workers. Similarly, the number of spindles employed per firm and the output volume vary substantially across firms. Descriptive statistics for Zaikabo firms are summarized in the second panel.

Table 1: Summary Statistics

a) Chinese firms					
VARIABLES	Mean	Std. dev.	Min.	Max.	Observations
Output (in yards)	26,395.52	33,345.89	730	601,000	403
Spindles	46,640.88	38,131.98	5,120	407,480	403
Capital assets (yen)	2,432,080	2,549,121	71,500	1.20e+07	403
Number of workers	2,680.191	2,491.412	270	30,800	403

The sample is restricted to firm-year observations with positive output. The estimation sample includes only firms with positive output and positive number of spindles, resulting in 403 observations. This includes 50 Chinese local firms in an unbalanced panel.

b) Zaikabo					
VARIABLES	Mean	Std. dev.	Min.	Max.	Observations
Spindles	29,764.16	26,001.23	1,020	338,000	375
Capital assets (yen)	2.55E+07	3.31E+07	2,400,000	1.60E+08	323
Founding year of plant	1904	15.31672	1887	1929	375
HQ in JPN	0.544	0.4987256	0	1	375

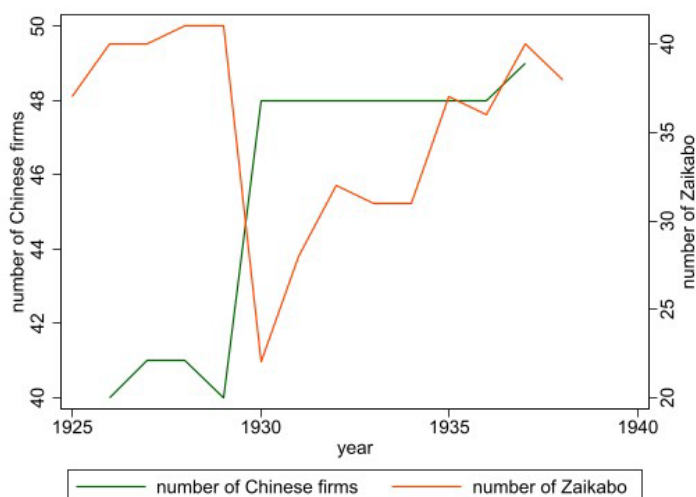
Note: The estimation sample is restricted to firm-year observations with a positive number of spindles. The time span is 1925-1938. The founding year is not Zaikabo's but the plant's.

Note that, in our data, the founding year is the year of starting operation of the plant rather than the Zaikabo. The oldest Zaikabo's plant dates back to 1887. As mentioned above (section 2), the first Zaikabo was founded in 1911 (Naigaimen, Co.), and most of the Zaikabo plants were taken over from various local Chinese firms (Minzokubo) and foreign-owned firms (Takamura, 1982: pp.75-78, pp. 116-123).¹⁸ Zaikabo often acquired cotton mills of exited firms or dormant/abandoned mills. The acquisition rather than green field FDI could save costs and time, which gave a large advantage to the Japanese firms in the Chinese market (Zhang, 2011: p.133). In addition, as pointed out in section 2, the entry of Japan to the Shanghai International Settlement was much later than that of European countries and the US. The major powers already dominated territories for business and manufacturing production. Furthermore, cotton mills were required to locate in riverside areas to transport raw materials and final products. As shown in Figure 1, cotton mills formed two clusters, Yangshupu district on the

¹⁸ For example, Shanghai Bouseki (the first Zaikabo) and Toka Bouseki took over the bankrupt Chinese local firms and Nikka Boushoku, Co. took over a British firm.

Huangpu river (East one in Figure 1) and Putuo district on the Suzhou Creek (West one). This implies that the Japanese FDI had almost no room for location choice, which reduces the selection bias in terms of geography inside Shanghai.

Figure 2: Trends in Active Zaikabo and Chinese Local Firms (Minzokubo) Over Time



Note: We define an active firm as one that has positive output in the respective year, according to our data.

Figure 2 illustrates the evolution of the number of active Zaikabo and local Chinese firms in our data. As is evident, Zaikabo were significantly affected by the economic crisis in 1930. The number of Zaikabo decreased from 41 in 1929 to 22 in 1930, but then recovered again to reach 38 by the year 1938. Over the same period, local Chinese firms experienced a steady increase in their numbers. This likely reflects several economic reforms (tariff, currency, and banking) by the Nationalist Government of China as mentioned above (section 2). In our data, the number of active local Chinese firms increased from 40 to 48 between 1926 and 1936.

4 Econometric Methodology

To investigate whether Japanese Zaikabo impacted the performance of Chinese cotton producers, we estimate sets of augmented production functions (as in, e.g., Aitken and Harrison, 1999; Javorcik, 2004; Lu et al., 2017). Output is measured as the total value of yarn produced. Firm-level inputs are the level of employment and the number of spindles used in the production process. We then analyse whether, conditional on these observable input factors, the presence of Zaikabo in the vicinity of the Chinese firm has any measurable impact on their output.

This idea for identifying FDI spillovers is implemented in the following production function equation:

$$y_{it} = \beta_1 s_{it} + \beta_2 l_{it} + \beta_3 pres_zaikabo_{it-1} + d_i + d_t + \varepsilon_{it} \quad (1)$$

where y is log output (in terms of yarn produced) by local Chinese firm i in year t , s is the log number of spindles, l is log employment, and d_i and d_t are full sets of firm and year dummies, respectively.¹⁹ $pres_zaikabo$ measures the lagged presence of Zaikabo in the vicinity of firm i , which will be defined in detail below. In this specification, the coefficient of interest β_3 measures, conditional on other variables in the model, the impact of changes in the presence of Japanese Cotton producers on the total factor productivity of Chinese firms. This is what we refer to as a “spillover effect”.

In order to get an idea of the possible mechanisms through which firms change productivity, we also look at the impact of the presence of Zaikabo on capital employed by Chinese firms. Specifically, we use the number of spindles as an outcome. The spillover estimate in this case indicates whether Chinese firms have adapted their stock of spindles (as an indicator of capital stock), which may suggest changes in production technology.

The standard approach to measuring FDI spillovers is generally based on a variable that measures the presence of foreign firms as the share of employment (or output) in foreign multinationals relative to total employment in firm i 's industry or in vertically related industries. To allow for a geographic dimension, this may be calculated for a geographic cluster, usually an administrative region or municipality (e.g., Girma et al., 2015, Aitken and Harrison, 1999). This is, of course, a fairly rough way of considering how the distance between local and foreign firms matters for spillovers through knowledge transfers. Given our detailed geographic data, we are able to follow a much more fine-grained approach.

We geo-reference all local Chinese firms and all Zaikabo $j \in J$ and compute the Euclidean distance for each pair $dist_{ij}$. We follow the literature on knowledge spillovers (e.g. Jaffe and Trajtenberg, 1996; Figueriredo et al., 2015) and construct a spillover term which given by

$$pres_zaikabo_{it} = \sum_{j=1}^J \frac{1}{(dist_{ij})^\alpha} \quad (4)$$

¹⁹ We note that the number of looms counted as spindle equivalent in our analysis follows the approach suggested by Kiyokawa (1974).

where α is set to unity in the benchmark specification and all Zaikabo are considered for each Chinese cotton producer. In other words, we allow for each Zaikabo to have a potential spillover effect, but this effect diminishes with distance.

Note that *pres_zaikabo* is firm *i* specific, as the distance between *i* and *j* varies across firms. This is an advantage over the spillover variables generally used in modern studies (as share of employment in foreign firms), which are industry-region, but not firm specific. For our spillover variable, time variation comes through new Zaikabos entering and existing Zaikabos becoming inactive. We only consider active Zaikabo for our purposes.

To look at more detail into the spatial characteristics of FDI spillovers, we also, in an extension to our baseline model, calculate spillover effects across different distance ranges. We then run alternative specifications with varying distance bands where we compute the spillover term for Zaikabo within the range *l* to *u* as $pres_zaikabo_{it}^{l,u} = \sum_{j=1}^J \frac{1}{(dist_{ij})^\alpha}$ for $l \leq dist < u$. Specifically, in what we report below we choose one distance band as between 0 and 10km, 0-20km and another as 0-30km. This reflects the finding in the literature that spillovers are generally geographically localised (e.g. Jaffe and Trajtenberg, 1996; Driffield and Girma, 2003; Figueriredo et al., 2015).

As these measures thus far are only based on number of Zaikabo, without giving regard to the size of the firm, we also use spillovers weighted by the size of the respective Zaikabo. This is measured in terms of the share of spindle of the respective Zaikabo in total production capacity in China (referred to as *pres_zaikabo_weighted_{it}*) as well as spillovers weighted by the size of the total number of spindles owned by the company to which the Zaikabo belongs to (i.e., summing over all China-based as well as non-China based plants of the company). Note that, in order to facilitate interpretation, we estimate standardized coefficients for the spillover terms so as to have comparable coefficients.

5 Estimation Results

Table 2 presents the results of estimating the models described above. The specification in column (1) estimates the Cobb-Douglas parameters for the two input coefficients. The parameters obtained have reasonable magnitudes, indicating a relatively high share of labor input, which is consistent with the characteristics of the cotton industry. Overall, this simple

and standard production function approach explains more than 70 percent of the variation in output.

Column (2) employs the distance-weighted measure of spillovers, *pres_zaikabo*, as defined in equation (4). The results indicate a statistically significant and positive coefficient for the distance-weighted spillover indicator. In other words, the findings suggest that cotton producers can increase their productivity when the number of Zaikabo firms in their vicinity rises. Given that the spillover variable is standardised, the point estimate gives the percentage change in productivity following a one-standard-deviation increase in the spillover variable. This is 15.1 percent in column (2), indicating that the estimated effect is not only statistically significant but also economically meaningful.

In columns (3) and (4) we use alternative spillover measures. Specifically, column (3) uses a size-weighted measure, *pres_zaikabo_weighted*, which accounts for firm sizes (proxied by the number of spindles). More specifically,

$$pres_zaikabo_weighted_{it}^{l,u} = \sum_{j=1}^J \frac{1}{(dist_{ij}) \times Spindles_j} \text{ for } l \leq dist < u \quad (5)$$

It thus allows for the fact that we might expect a greater spillover potential from larger Zaikabos. Reassuringly, the results found using this size-weighted measure are very similar to the ones in column (2), in terms of sign, statistical significance, and magnitude of the estimated effect. Indeed, the coefficient estimate in column (3) compared to column (2) may suggest that the size of Zaikabo firms plays a role, as the estimates become somewhat more precise with the size-weighted measure of spillovers. It seems a reasonable assumption that larger firms generate more potential for spillovers. If spillovers happen through the movement of workers from Zaikabo to Chinese local firms (as alluded to above, section 2), then larger firms, with a larger number of workers, may generate more spillovers than smaller firms.

In column (4), we replace the size weights of individual Zaikabo firms with a measure based on the size of their parent firm, specifically the total number of spindles across all plants owned by the respective parent firm of the Zaikabo. Once again, we find strong evidence of spillovers, with the magnitude of the effects increasing slightly. In subsequent analyses, we adopt the size weights, accounting for the respective Zaikabo firm's size as our preferred measure for spillovers.

Table 2: FDI spillover effects

	(1)	(2)	(3)	(4)	(5)
	Log Output			Log spindles	
<i>pres_zaikabo</i>		0.151** (0.0686)			
<i>pres_zaikabo_weighted</i>			0.126*** (0.0473)		0.0833** (0.0397)
<i>pres_zaikabo_weighted_HQ</i>				0.330** (0.159)	
log workers	0.484*** (0.0991)	0.489*** (0.0987)	0.486*** (0.0971)	0.487*** (0.079)	0.360*** (0.0791)
log spindles	0.210** (0.0996)	0.202** (0.0996)	0.192* (0.0980)	0.192** (0.077)	
Observations	403	403	403	403	403
R-squared	0.764	0.765	0.767	0.767	0.883

Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1 . All columns include firm fixed effects and time fixed effects. Standardized coefficients are reported the for the spillover terms *pres_zaikabo* (inverse distance weighted sum of active Zaikabo), *pres_zaikabo_weight* (inverse distance weighted sum of active Zaikabo weighted by the size of the Zaikabo), and *pres_zaikabo_weight_HQ* (inverse distance weighted sum of active Zaikabo weighted by the size of the firm the Zaikabo belongs to)

Finally, in column (5), we look at one potential mechanism of how the productivity improvement that local firms experience may come about. Specifically, we look at whether local firms increase their capital stock. To do so, we shift the outcome variable to the number of spindles used by Chinese firms. The results suggest that the positive productivity effect of Zaikabo firms in proximity to Chinese firms may partly stem from an increase in the number of spindles used by Chinese producers. This indicates that Chinese firms may be upgrading their technology and shifting towards greater capital usage.

The literature suggests that it is reasonable to assume that spillovers are strongest if the foreign firm (Zaikabo in our context) is in the immediate geographic vicinity and that they peter out over longer distances (e.g. Jaffe and Trajtenberg, 1996; Driffield and Girma, 2003). This may be particularly relevant in the historic context we consider here, as information was arguably far less mobile and transferable than it is in today's age of modern communication technologies. In order to allow for the role of distance, we now calculate spillover measures for three distance bands: the first being within 0 to 10km from the local firm, the second 0 to 20km, and the third 0 to 30km. The results are reported in columns (1) to (3) in Table 3.

We report estimation results, using the size-weighted spillover variables as in Table 2, column (3), in all specifications of Table 3. The results in columns (1) to (3) clearly show that distance indeed matters: we only find statistically significant spillover effects from Zaikabo located within 0 to about 20km away from the local firms. The coefficient magnitudes monotonically decrease with increasing the distance bands. While effects are still of economic relevance at 30km distance band, they are not statistically significant anymore, and they further decline when increasing the distances to Zaikabo further away than 30km (these results are not reported to save space).

Table 3: Distance thresholds of spillover effects

	(1)	(2)	(3)
	Distance X=10km	Distance X=20km	Distance X=30km
<i>pres_zaikabo_weighted</i> distance 0 - X km	0.155** (0.0663)	0.135** (0.0665)	0.106 (0.0674)
<i>pres_zaikabo_weighted</i> distance > X km	-0.0587 (0.0613)	0.0215 (0.0581)	-0.0406 (0.0595)
log workers	0.492*** (0.0798)	0.488*** (0.0801)	0.480*** (0.0801)
log spindles	0.190** (0.0780)	0.189** (0.0786)	0.199** (0.0786)
Observations	403	403	403
R-squared	0.767	0.767	0.767

Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1 . All columns include firm fixed effects and time fixed effects. Standardized coefficients are reported the for the spillover terms *pres_zaikabo_weighted* (inverse distance weighted sum of active Zaikabo weighted by the size of the Zaikabo).

We now turn to look at a number of other issues related to heterogeneity of the spillover effects, the results of which are reported in Table 4. As we saw in the map produced in Section 3 (Figure 1), Chinese local firms, as well as Zaikabo, are heavily concentrated in the city of Shanghai. We therefore check whether this city is driving our results, by allowing the spillover effect to be different for local firms located in Shanghai and those in the rest of the country. This is done by interacting the spillover variable with an indicator equal to one if the local firm is in Shanghai. These results are reported in Table 4, where we always estimated the baseline specification as presented in column 3 of Table 2 and added an interaction term for the corresponding heterogeneity. Table 5 reports only the main effect of the spillover for the reference and the comparison group. With regard to the location in China, the results in the

firm. Firm size, measured by the capital assets of Chinese firms, appears to be relevant insofar as the point estimates are larger for above-average-sized firms. However, the estimates are also statistically significant for smaller firms, and the confidence intervals of the spillover effects overlap across both size groups.

Somewhat surprisingly, we find that the spillover effects are more pronounced for single-plant firms. One explanation could be due to the pre-existing knowledge capacity that is arguably less pronounced in single-plant than in multi-plant firms. This could be explained by the better potential to take up information and knowledge from Zaikabo. With regard to the age of the local Chinese firms, we observe little heterogeneity in the magnitude of the effects, but the statistical significance is higher for the younger firms. A significant effect pertains to the ownership type of local firms: firms with some foreign-ownership participation tend to see more pronounced spillover effects. Note that the size-weighted spillover effects are still significant at the 15 percent level for non-foreign firms, and the spillovers without size weights remain significant at the 5 percent level with a magnitude of 0.152 for non-foreign firms. Hence, the spillovers were not limited to firms with some foreign ownership but they were clearly more pronounced.

As noted in Section 2, relations between Japan and China deteriorated rapidly in the 1930s, particularly after the 1931 *Mukden Incident* (*Manchurian Incident*). In such a hostile political environment, one might expect interactions between firms—and their workers—to be limited, thereby restricting potential spillovers. To examine whether this is reflected in our data, we interact our spillover variable with a dummy variable that equals one for years after 1931 and zero otherwise. This allows us to assess whether the nature of spillovers changed following the *Mukden/Manchurian Incident*. The results, reported in the last row of the firms panel in Table 5, indicate that the coefficient turns negative when the sample is restricted to post-1931 observations. However, introducing an interaction term to examine group-specific effects reduces within-firm variation, which is crucial for identification in fixed-effects specifications. As a result, the effect remains positive in earlier years but ceases to be statistically significant. These findings suggest that Chinese firms adapted their production technology and no longer benefited in terms of output once spillovers from *Zaikabo* diminished. This is consistent with the above-mentioned evidence that *Zaikabo* shifted their production to Manchuria and North China to prevent a boycott of Japanese goods and intensified the anti-Japanese movement after the Mukden incident.

We also examine the implications for spillovers of heterogeneities among Japanese *Zaikabo*. *Zaikabo* firms had substantial heterogeneity in ownership, management system, products, firm organization, process/background/history/timing of investment in China (Takamura, 1982). Therefore, not all types of *Zaikabo* might have the same potential for spillovers, and this is what we investigate here. We first consider whether the *Zaikabo* reports its headquarters to be in China or in Japan. Some firms (e.g. Naigaimen) headquartered in Japan and established branch offices and plants in China, while some established affiliated companies headquartered in China with parent firms in Japan and some (e.g. Shanghai Bouseki) were headquartered in China and established branch offices in Japan. The idea is that the potential for spillovers may be higher if the HQ is not located in Japan, as this may indicate that firms are more deeply invested in China and are more internationally oriented and thus generate stronger spillovers. We also consider whether the HQ is located in Manchuria, with the same reasoning applying.

To look at this, we calculate two spillover variables as in equation (4), one based on *Zaikabo* with HQ in Japan and the other without. We then include both variables in the productivity estimation and report both coefficients in Table 5. We proceed similarly in the case of HQs in Manchuria. Our findings suggest that spillover effects are most pronounced for *Zaikabo* headquartered outside both Japan and Manchuria, in line with our expectations.

Furthermore, we also investigate whether the spillover potential differs depending on the size and the age of the *Zaikabo*. We would expect a higher potential for large and old firms to generate spillovers. Larger firms have a higher absolute number of employees, which can potentially move to local firms and transfer knowledge, thus facilitating job-hopping as discussed in Section 2. Also, large and older firms may have a more established knowledge about the industry and the competitive environment, knowledge that can be usefully transferred to local firms. To check this aspect of heterogeneity, we first distinguish firms by employment size into whether they are larger than the lowest percentile of the size distribution or not. We also use an alternative measure defined using the median. Firms are also classified by age using the median age.

Results, shown in the bottom rows of Table 5, indicate that, regarding firm size, firms on either side of the median exhibit similar spillover effects, whereas very small firms—those in the first quartile of the size distribution—do not display significant spillovers. Similarly, older *Zaikabo*

generate stronger spillover effects. These findings suggest that a minimum threshold of firm size and age is necessary for a *Zaikabo* to possess the human capital or knowledge required to contribute meaningfully to positive spillovers for local Chinese firms.

Table 5: Heterogeneity of spillover effects depending on *Zaikabo*

VARIABLES	(1)	(2)
	I=0 (No)	I=1 (Yes)
<i>Headquarter in Japan</i>	0.154*** (0.060)	0.023 (0.063)
<i>Headquarter in Manchuria</i>	0.159** (0.066)	0.091 (0.073)
<i>Firm size > 1. Quartile</i>	0.026 (0.039)	0.146** (0.070)
<i>Firm size > Median</i>	0.271*** (0.090)	0.243*** (0.072)
<i>Age > Median</i>	-0.013 (0.059)	0.159*** (0.075)

Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Standardized coefficients for the interaction terms are reported. All specifications include the same control variable as the main specification in column 3 of Table 2.

6 Conclusions

In this paper, we use comprehensive historic firmlevel data to estimate productivity spillovers from Japanese textile companies' affiliates in China (*Zaikabo*) to local cotton producers in China. We have access to very detailed information on all *Zaikabo* as well as local textile firms during the period 1925 until 1938, a period of tumultuous relationships between Japan and China. We geo-localized and digitized this information in order to allow us to pinpoint the exact location of the firms. This enables us to also capture the role of distance in facilitating productivity spillovers.

Our results provide clear evidence for positive spillovers from the Japanese foreign affiliates (*Zaikabo*) to local Chinese firms. They are able to improve their overall productivity and this goes hand-in-hand with a change in production technology towards greater use of capital / spindles. Our paper is thus the first to provide evidence for such spillovers in a historical context. We also find that spillovers are very localised, being strongest within a radius of up to 10km around the *Zaikabo* and then petering out with further distance. These extremely

localised spillover results may also explain why much of the “spillovers literature” generally fails to find strong evidence for spillovers in the same industry: these studies are not able to zoom in on the geographic dimension in the same detail as we are able to.

At another cut at the geographic dimension of our data, we find that evidence for spillovers is particularly strong for firms in Shanghai. This reflects the importance of Shanghai as the centre for Cotton textile production at the time.

Overall, our paper shows that FDI spillovers matter also in a historic context. From today’s perspective, our paper, by focusing on a very distinct sector with a fairly homogeneous product, provides a novel identification of spillovers, as we can abstract from industry heterogeneity and to a large extent also endogenous plant location, that plague many modern studies using data for the whole manufacturing sector (e.g., Girma et al. 2015 or Lu et al. 2017 for China). We are therefore able to clearly pinpoint the source of spillovers and the geographical dimension to it. Our historical approach hence shows that horizontal spillovers can be important.

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Scaling up sanitation: Evidence from an RCT in Indonesia[☆]Lisa Cameron^{a,*}, Susan Olivia^b, Manisha Shah^c^a University of Melbourne, Australia^b University of Waikato, New Zealand^c University of California, Los Angeles and NBER, USA

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ABSTRACT

We investigate the impacts of a widely used sanitation intervention, Community-Led Total Sanitation, which was implemented at scale across rural areas of Indonesia with a randomized controlled trial to evaluate its effectiveness. The program resulted in modest increases in toilet construction, decreased community tolerance of open defecation and reduced roundworm infestations in children. However, there was no impact on anemia, height or weight. We find important heterogeneity along three dimensions: (1) poverty—poorer households are limited in their ability to improve sanitation; (2) implementer identity—scale up involves local governments taking over implementation from World Bank contractors yet no sanitation and health benefits accrue in villages with local government implementation; and (3) initial levels of social capital—villages with high initial social capital built toilets whereas the community-led approach was counterproductive in low social capital villages with fewer toilets being built.

1. Introduction

It is estimated that about 1.1 billion people worldwide practice open defecation as a result of lack of access to sanitation facilities. Diseases caused by open defecation are preventable and disproportionately affect the poor. Millions of people contract fecal-borne diseases, most commonly diarrhea and intestinal worms, with an estimated 1.7 million people dying each year because of unsafe water, hygiene and sanitation practices (WHO/UNICEF, 2010). In Indonesia 110 million people lack access to proper sanitation and 63 million practice open defecation (WHO/UNICEF, 2012). Two of the four main causes of death for children under five in Indonesia (diarrhea and typhoid) are fecal-borne illnesses linked directly to inadequate water supply, sanitation, and hygiene issues (Ministry of Health, 2002). About 11 percent of Indonesian children have diarrhea in any two-week period and it has been estimated that more than 33,000 die each year from diarrhea (Curtis, 2004). By reducing normal food consumption and nutrient absorption, diarrheal diseases and intestinal worms are also a significant cause of malnutrition, leading to impaired physical growth

(Guerrant et al., 1999), reduced resistance to infection (Baqui et al., 1993), and long-term gastrointestinal disorders (Schneider et al., 1978). Inadequate sanitation is associated not only with adverse health effects, but also with significant economic losses. Inadequate sanitation and poor hygiene in Indonesia is estimated to cost approximately US\$6.3 billion, or more than 2.4 percent of the country's gross domestic product (GDP) (Napitupulu and Hutton, 2008).

Community-Led Total Sanitation (CLTS) is a program being widely implemented in more than 60 countries throughout Asia, Africa, Latin America, the Pacific and the Middle East to address the sanitation burden (Wells and Sijbesma, 2012). CLTS aims to create demand for sanitation by facilitating graphic, shame-inducing community discussions of the negative health consequences of existing sanitation practices, rather than through the more traditional approach of providing sanitation hardware or subsidies. The Water and Sanitation Program (WSP) of the World Bank is facilitating the implementation of CLTS widely. As part of a learning agenda to address the burdens associated with poor sanitation, the Bill and Melinda Gates Foundation funded randomized controlled trials (RCTs) of sanitation and

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hygiene interventions in seven locations around the world.¹ This paper presents the results of the Indonesian evaluation. We report the impact of CLTS on outcomes of interest along the causal chain as improvements in sanitation have the potential to lead to a decrease in parasitic infestations, a decrease in anemia, and an increase in weight and height for young children. We rely on objective measures of impact—physical inspection of sanitation facilities, blood and fecal samples, and physical anthropometric measures. The evaluation results show that CLTS modestly increased the rate of toilet construction, decreased community tolerance of open defecation, and reduced the prevalence of roundworm infestation. There is no discernible impact on the other health measures and when we generate an overall health index including roundworm infestation, anemia, height and weight, there is no significant treatment effect. Allowing for heterogeneous treatment effects shows that the program is less effective among poorer households.

In addition to poverty status, we examine two other sources of heterogeneity in program impact. An important component of the intervention is that it sought to create a large-scale sustainable program across the country.² WSP contracted resource agencies to implement the program in a set number of villages. The resource agencies were also contracted to train local government staff to implement the program themselves (partly through the observation of one or more of the resource agencies' implementations). During the phase of the project we evaluate, the resource agencies and local governments were implementing the program simultaneously. Approximately half of our treatment villages were treated by the resource agencies (RA) and the other half by local government (LG). This allows us to examine how program impact varies with the identity of the implementer and hence evaluate the scale-up process. As villages were not randomly allocated to implementing teams, the estimates examining this form of heterogeneity rely on the assumption that there are no unobserved differences between RA and LG villages which could be causing a differential impact. Discussions with WSP suggest there was no systematic process of assignment and tests of household and village baseline characteristics by implementer status show no significant differences. We find that while statistically significant program impacts are observed in communities where the program was implemented by a resource agency, local government implementation produced no discernible

benefits.³

Second, we examine the role of social capital. As its name suggests, CLTS is a participatory development project in which facilitators are sent to villages to initiate a community analysis of existing sanitation practices and a discussion of the negative health consequences of such practices. The community actively participates in the facilitated meeting and is then left to forge its own plan to improve village sanitation with only limited follow-up support and monitoring from the program. Given CLTS' emphasis on community involvement, one might expect that it would function best in communities with high pre-existing levels of social capital. We define social capital as the norms and networks that enable collective action.⁴ Since social capital is not randomly allocated to villages, estimating these heterogeneous treatment effects assumes that there are no unobserved variables which differ between low and high social capital communities that could be driving the results. While we cannot rule out that this might be the case, the stability of the estimates when a wide array of control variables are included suggests it is unlikely that the estimates are biased due to omitted variables (Altonji et al., 2005).

We find that baseline social capital is an important determinant of program effectiveness. High levels of community participation at baseline are strongly associated with increased toilet construction in treatment communities. However, in communities with low levels of social capital at baseline, significantly fewer toilets are constructed. We present results consistent with villages with higher initial community participation being more able to align community members' objectives with those of the program through the use of social sanctions. Our results thus constitute a caution about using participatory approaches in low social capital settings. At the very least they suggest that more intense involvement of project facilitators and general project support may be warranted in locations with demonstrably low social capital.

The paper proceeds as follows. Section 2 provides details of the intervention, the experimental design and data collection. Sections 3 and 4 explain the estimation strategy and present the main impact evaluation results. Section 5 examines heterogeneous treatment effects. Section 6 concludes.

2. Intervention and study design

CLTS was initially developed in Bangladesh in 1999 by sanitation practitioner, Kamal Kar. It is now being widely implemented in more than 60 countries around the world (Wells and Sijbesma, 2012), having

¹ CLTS was implemented in Indonesia, Tanzania and two locations in India. Handwashing interventions were implemented in Peru, Vietnam, and Tanzania. Results from the CLTS evaluations are reported in Patil et al. (2014) and Briceno et al. (2017), for India and Tanzania respectively. Both find increases in the ownership of improved sanitation (15 percentage point increase – from 50 to 65% – in Tanzania and 19 percentage points in India – from 22 to 41%). The Indian program is called the Total Sanitation Campaign and pairs the CLTS approach with toilet construction subsidies. An additional study, Pickering et al. (2015), conducted a clustered RCT of CLTS in Mali and finds increased toilet construction, no change in diarrhea prevalence but improvements in child growth.

² Recently there has been a push for a greater use of randomized experiments “at scale” (Davis et al., 2017; Muralidharan and Niehaus, 2017).

³ We are aware of only a small number of studies that conduct rigorous quantitative evaluations of the scaling up process, all of which highlight the many unforeseen difficulties in scaling up projects and the need to carefully evaluate the scale up process. Bold et al. (2013) find that an educational intervention increased student test scores when implemented by an NGO in Western Kenya but failed to increase scores when replicated at scale by the government. Grossman, Humphreys and Sacramont-Lutz (2015) finds that the high take-up by marginalized populations of new low-cost technology that allows constituents to engage with their local politicians could not be replicated when scaled to comprehensively cover half the country. A more promising outcome is reported in Banerjee et al. (2017) and Banerjee et al. (2016) who build on knowledge gained through previous failed attempts to effectively scale up the “Teaching at the Right Level” program. Further, Duncan and Magnuson (2013), in a review of the impacts of pre-school programs, note that the results from programs implemented for large and representative populations are generally much smaller than those found for small-scale pilot programs.

⁴ Many related definitions of social capital exist which generally incorporate preferences that inform interactions amongst individuals in a pro-social manner, such as altruism, trust and reciprocity; and/or a set of underlying community networks that can be used by individuals for private or public benefit (Durlauf and Fafchamps, 2005). See also Putnam et al. (1993), Coleman (1988) and Grootaert et al. (2002).

been adopted by many international NGOs (for example, Plan International, UNICEF, Care, World Vision) and the World Bank. Governments are increasingly taking the lead in scaling up CLTS with many having adopted CLTS as national policy. CLTS is viewed by many in the water and sanitation sector as the most promising approach to improving sanitation currently available.

The program is a community-led approach that focuses on creating demand for sanitation, in contrast to the traditional approach of supplying sanitation hardware (Sah and Negussie, 2009). CLTS facilitators are sent to villages to initiate a community analysis of existing sanitation practices and a discussion of the negative health consequences of such practices. The community actively participates in the facilitated meeting and is then left to forge its own plan to improve village sanitation with only limited follow-up support and monitoring from the program. These discussions, or “triggerings” are held in public places and are open to all. They involve a “walk of shame,” during which the facilitator helps people analyze how fecal contamination spreads from exposed excreta to their living environments and food and drinking water. A map of the village is drawn on the ground and villagers are asked to indicate where they live, where they defecate, and the routes they take there and back. This illustrates that everyone is ingesting small amounts of each other’s feces which is intended to lead to individual and collective decisions to improve community health by becoming an open defecation free (ODF) community. ODF status is verified by local government agencies and community members.⁵

In contrast to other approaches that have been used widely in the past in Indonesia and elsewhere, no funding for infrastructure or subsidies of any kind is provided. CLTS founders believe that CLTS is less effective when subsidies are available (Kar and Pasteur, 2005). They argue that the existence of subsidies causes people to postpone investing in sanitation in the hope that they will receive a subsidy and that subsidies instill a culture of dependency rather than self-determination. The lack of subsidies also makes the program much less expensive and savings can be utilized to spread and scale up the { } program.

2.1. Randomization design and data collection

In Indonesia CLTS was rolled out across all 29 rural districts in the province of East Java. East Java is Indonesia’s second most populous province with approximately 38 million residents. Eight of the 29 rural districts in East Java were involved in the impact evaluation.⁶ In each district ten villages were randomly selected to participate in the impact evaluation as treatment villages and ten were randomly selected to act as control villages. The district offices were free to implement the program in other villages, other than the control villages. Most district offices implemented the program in 40–70 villages, with the program

⁵ In Indonesia the program is called Total Sanitation and Sanitation Marketing (TSSM) or in Indonesian, Sanitasi Total & Pemasaran Sanitasi (SToPS). It consists of a CLTS demand-side component and also a supply-side component which seeks to support the development of the local sanitation market. The supply-side component was however not well developed at the time of the evaluation (Cameron et al., 2013). For more information on CLTS see <http://www.communityledtotalsanitation.org/page/clts-approach>.

⁶ East Java’s 29 rural districts were divided into three groups: Phase 1 districts received the program first, Phase 2 districts received it next, and Phase 3 districts received it last. The evaluation was conducted in Phase 2 districts. Phase 2 was chosen largely on the basis of timing. Evaluating the program in Phase 2 districts provided sufficient time for the baseline survey to be conducted prior to program implementation. Many of the start-up issues confronted in Phase 1 were sorted out by Phase 2 so the evaluation provides an impact estimate which is more representative of what could be expected from a national scaling up of the program following such large-scale piloting.

intending to reach a total of 1.4 million people across all rural districts in East Java (Cameron and Shah, 2010). Randomization was conducted at the village level, stratified by sub-district.⁷ There was only partial compliance with the randomization assignment. Of the 80 treatment villages, the endline survey data reports that 53 villages (66 percent) were triggered and 13.8 percent of the control villages were exposed to the program. Non-compliance was largely a result of district governments changing some of their target communities after the randomization plan had been agreed upon. Program administrative data collected as part of the CLTS program reports a higher percentage of treatment villages (83 percent) and a smaller percentage of control villages (4%) received the treatment. Below we estimate the average treatment effect across villages that were assigned to treatment, that is Intention-to-Treat (ITT) estimates.

Two waves of household data were collected. The baseline survey was conducted just prior to program implementation in August–September 2008. Within each village, approximately thirteen households were randomly selected to be surveyed. The endline data collection was conducted approximately 24 months later, between November 2010 and February 2011. The surveys collected a wide variety of information on the households including demographic information, a detailed sanitation module (including physical observations of household sanitation facilities which are used to verify household reports), and a child health module (including fecal samples to allow testing for parasitic infestations, blood tests for anemia, and anthropometric measurements). Fecal samples were collected by leaving a stool specimen container containing a preservative (formalin) with the child’s carer and requesting that he/she deposit approximately 5 g of the child’s feces in the container the next time the child defecates. The specimen was collected the next day. Preserved stool samples were sent to the public health office in Yogyakarta for analysis using the Kato-Katz technique (Katz et al., 1972). The fecal samples were tested for hookworm, roundworm, and whipworm. Prevalence of hookworm and whipworm was extremely low in the sample, with less than 1% testing positive for either of these types of worms, so we only use roundworm in the analysis. Hemoglobin was measured using HemoCues on a pinprick blood sample collected by trained field staff. Height and weight were measured using the standardized protocols developed by the Demographic and Health Surveys, see ICF International (2012).

To enable an examination of impacts on child health, households with children under the age of two were prioritized, with all surveyed households required to have at least one child under the age of five at baseline. Community level demographic data and information on infrastructure were also collected. In addition, a social capital module was conducted. Budgetary considerations restricted the social capital module to a randomly chosen six of these eight districts.⁸ Program administrative data identifies the implementing agency in each village.

Our total sample consists of approximately 2000 households spread across 160 rural villages in eight districts, with data on social capital available for approximately 1600 households spread across 120 villages in six of these districts.

3. Empirical strategy

Our empirical approach is to present ITT estimates of program impact on the outcomes of interest. This is done by estimating equa-

⁷ The districts are spread across East Java and include Probolinggo, Bonodowoso, Situbondo, Banyuwangi, Ngawi, Madiun, Jombang and Blitar. For a more in-depth description of the geographic context, see Cameron and Shah (2015). For further detail on the randomization design and process see Cameron and Shah (2010).

⁸ These six districts are Blitar, Bonodowoso, Jombang, Ngawi, Probolinggo, and Situbondo.

tion (1) below:

$$Y_{ij} = \alpha + \beta_1 T_j + \gamma X_{ij} + \delta_K + \epsilon_{ij} \quad (1)$$

where Y_{ij} is the outcome measure for household i in village j ; T_j is the treatment dummy, which equals 1 for households in the treatment group, and 0 otherwise; δ_K is a set of sub-district (*kecamatan*) dummy variables which are included because the randomization was stratified at this level. The sub-district effects also control for any differences in implementation across the eight districts. In some specifications, we also include a vector of household and village characteristics (X_{ij}) as additional right-hand side control variables. The household variables are household size, the household head's age and educational attainment, household composition, log per capita household income, eligibility for low income support and dwelling characteristics; and the village variables are village population, village land area, the percentage of the village which is Muslim, whether there is a paved road to the nearest city, average years of education of household heads, whether a river flows through the village, and the percentage of households in the village who open defecated at baseline. ϵ_{ij} is the error term, and all specifications cluster the standard errors at the village level.

The causal average treatment effect is given by β_1 if the randomization was effective. Table 1 uses the 2008 baseline survey data to compare characteristics of treatment and control groups. It shows that the means of a range of variables are similar in magnitude for the two groups and we cannot reject that they are equal for most of the variables. For the key outcome variables (sanitation, child health outcomes, attitudes toward open defecation), balance is achieved. The demographic and socio-economic characteristics are also similar across treatment and control groups. The baseline report provides tests of balance on a more extensive set of variables (Cameron and Shah, 2010).

When examining heterogeneity of impact, we include interactions of T_j with the relevant variables – poverty status of the household; whether the village was assigned to be treated by a resource agency (RA) or by the local government (LG); and baseline social capital. We discuss the issue of balance with respect to the relevant sub-samples in the analyses below.

4. Empirical results

Table 2 reports the ITT estimates for the main outcomes of interest—toilet construction, attitudes toward open defecation, knowledge of the causes of diarrhea and child health outcomes. Control variables are included in these specifications. Table A1 in the appendix reports results when control variables are not included. The results are similar.

We first examine whether CLTS treatment was successful in stimulating demand for sanitation (column 1). Households report whether they built a toilet between baseline and endline. This report is verified at the end of the interview by an inspection of the household's sanitation facilities. Table 2 shows that treatment increases toilet construction by 2.4 percentage points. This constitutes a 19 percent increase in toilet construction relative to control communities.

CLTS is hypothesized to stimulate demand for sanitation by inducing shame associated with open defecation. It also imparts information on the negative health consequences of poor sanitation. Column 3 reports the results of whether treatment impacts a measure reflecting the degree to which the respondent agrees (disagrees) with negative (positive) views of open defecation. There is a small decrease in the community's tolerance of open defecation in treatment communities relative to control communities. The program does not affect knowledge of the causes of diarrhea (unclean water, not washing hands, open defecation, etc.) which may be due to knowledge being quite high already, with

the mean score in control communities being 4.9 out of 6 (column 5).⁹

The ultimate aim of CLTS is to improve sanitation so as to improve community health, particularly child health. We examine the health impacts along the causal chain from roundworm infestation, to hemoglobin blood concentrations (with low hemoglobin indicating anemia) to weight and height z-scores (columns 6–11). While columns 1–5 are estimated at the household level, the health regressions are estimated at the child level. The sample of children is those aged 0 to 5 at endline.

Column 6 of Table 2 shows that treatment is associated with an approximately 46% decrease in roundworm infestation. This is a large decrease, which is surprising given the relatively modest increase in access to sanitation. Such a large decrease in roundworms would have the potential to have significant impacts on nutritional intake and so might be expected to be reflected in anemia, weight or height gains. However, we do not find any significant treatment effects on hemoglobin concentrations, weight or height z-scores.

In column 11 we generate an overall health index. Following Kling et al. (2007) and Casey et al. (2012), we construct the index by orienting all variables so that the positive direction indicates a better outcome; demeaning all re-oriented outcomes and dividing each variable by the control group standard deviation; and calculating the average of these variables. While the coefficient on treatment is positively signed, it is not statistically significant (column 11).

5. Heterogeneous treatment effects

5.1. Scale up

CLTS forms part of the Indonesian government's national strategy to improve environmental and health outcomes in rural areas. Ensuring sustainability of the project by embedding implementation in district governments was the key element of the scale up strategy. WSP followed the World Health Organization's documented steps for developing a successful scale up strategy which were formed on the basis of evidence gathered over years of experience in scaling up public health

⁹ For the attitudinal measure the respondent is asked whether s/he agrees, strongly agrees, disagrees or strongly disagrees with: Having a toilet of our own will stop my family becoming a target of gossip; Sanitation facilities in this village improves the community as there is no longer environmental pollution; Most people that I know defecate in a toilet; It is OK to defecate in the open as our ancestors did; Having our own toilet will reduce the likelihood of family members getting diarrhea; It is OK to defecate in the river as others do it; It is acceptable for children to defecate in the open; It is acceptable to defecate in the open if you don't have a toilet; People who defecate in the open will not be accepted by the community. The aggregated attitudinal measure is the sum of responses to these 9 questions about attitudes toward open defecation (oriented so a higher score indicates greater intolerance of open defecation). 45 is the maximum score possible and is the highest level of intolerance while 9 is the minimum score possible and reflects total acceptance of open defecation. For the knowledge of the causes of diarrhea the caregiver is asked to indicate whether the following activities cause diarrhea: drinking dirty water; using dirty latrines; other people defecating in the river; other people defecating in another open space (yard/rice field/beach/etc); not washing hands with water; not washing hands with soap. The aggregated knowledge measure is a score out of 6 with a score of 6 indicating that the respondent got all of the questions correct.

Table 1
Summary statistics: Means and tests of balance.

Variable	All Villages			Treatment Villages		
	Treatment	Control	p-value	RA Treatment	LG Treatment	p-value
Panel A. Baseline Household Characteristics:						
Do not have own sanitation facility	0.52	0.48	0.44	0.52	0.52	0.95
Open defecate	0.40	0.40	0.93	0.40	0.39	0.89
Access to Unimproved Sanitation	0.11	0.12	0.76	0.12	0.10	0.66
Access to Improved Sanitation	0.49	0.48	0.74	0.48	0.51	0.69
Wash hands after defecation	0.99	0.99	0.20	0.99	0.99	0.80
Access to piped water	0.05	0.08	0.39	0.05	0.05	1.00
Intolerance of open defecation	33.1	32.9	0.70	33.6	32.6	0.19
Believes open defecation causes diarrhea	0.69	0.67	0.50	0.69	0.69	0.89
Knowledge of causes of diarrhea	4.73	4.73	0.94	4.81	4.66	0.39
Household head's age	40.4	40.4	0.99	40.2	40.5	0.68
Household head male	0.95	0.96	0.31	0.95	0.94	0.52
Household head's educational attainment:						
Elementary	0.52	0.48	0.27	0.53	0.51	0.68
Lower Secondary	0.20	0.21	0.46	0.19	0.21	0.68
Upper Secondary	0.18	0.21	0.28	0.19	0.18	0.74
Tertiary	0.04	0.04	0.78	0.04	0.05	0.76
Household size	4.92	4.82	0.29	5.00	4.83	0.20
No. children aged 0-5	1.15	1.12	0.07*	1.15	1.15	0.85
No. children aged 6-10	0.33	0.31	0.45	0.32	0.34	0.56
No. children aged 11-17	0.39	0.40	0.65	0.39	0.39	0.91
Per capita household income	2.81	3.02	0.37	2.84	2.79	0.86
House has a dirt floor	0.22	0.23	0.91	0.25	0.20	0.33
House has a tiled floor	0.05	0.07	0.23	0.05	0.05	0.83
House has walls of brick or wood	0.88	0.89	0.86	0.87	0.90	0.30
Household uses wood as a cooking fuel	0.57	0.52	0.21	0.56	0.59	0.57
Cash transfer program (BLT) recipient	0.24	0.23	0.72	0.26	0.23	0.36
Max Observations:	922	936		460	462	
Panel B. Baseline Child Characteristics:						
Age in months	11.8	12.1	0.30	12.1	11.6	0.24
Male	0.50	0.52	0.44	0.49	0.51	0.48
Hemoglobin (g/l)	101.5	101.8	0.81	100.7	102.4	0.22
Weight (kgs)	8.2	8.3	0.61	8.2	8.3	0.49
Height (cms)	71.2	71.6	0.34	71.4	71.1	0.58
Max Observations:	946	940		471	475	
Panel C. Baseline Village Characteristics:						
Village population	1042	1300	0.14	824	1249	0.02**
Paved road to nearest city	0.94	0.95	0.73	0.95	0.93	0.69
% of the village population that are muslim	97.6	95.4	0.19	97.4	97.9	0.67
River runs through village	0.72	0.71	0.86	0.72	0.73	0.89
% of households in village that open defecate	33.6	36.4	0.52	32.8	34.5	0.78
Average years of education of household heads	10.3	10.5	0.33	10.2	10.3	0.85
Village land area	42,816	34,930	0.70	42,377	43,234	0.98
Social capital index at BL	0.01	-0.02	0.61	-0.02	0.04	0.46
Max Observations:	80	80		39	41	

Note: These are summary statistics (means) using the baseline data. RA (LG) Treatment indicates villages which were assigned to implementation by a resource agency (local government). Information on roundworm prevalence and intolerance of open defecation is not available at baseline. The p-values are generated from tests of statistical difference between treatment and control communities.

interventions (WHO, 2010).¹⁰

The scale up model used was the widely-employed “Training of Trainers” (Binswanger and Nguyen, 2004). WSP trained staff at resource agencies (RA) which had successfully bid for the work and then these resource agencies trained local government (LG) officials in

CLTS with the local government then taking over and scaling up the program to all villages (Rosenzweig and Kopitopoulos, 2010). A portion of this training was done by demonstration or “learning-by-doing”, as LG officials observed RAs triggering some villages. This process took place at the time of the RCT. Amongst the treatment villages in our sample, we have 39 villages triggered by the RAs and 41 villages triggered by the LGs.

There are a number of reasons why program impacts may differ when conducted at scale:

1. *Demographic context.* Different characteristics of target populations when at scale.
2. *Design effects.* Differences in program design that are necessary to bring the program to scale.
3. *Scale effects.* General equilibrium effects associated with the scale of the project, including spillover effects.

¹⁰ Owing to CLTS forming part of a Gates Foundation global learning agenda, the CLTS program itself and the scale up strategy are unusually well-documented. For example, see Kar and Chambers (2008); Rosenzweig and Kopitopoulos (2010); Mukherjee (2009, 2011); Pinto (2013); for a global discussion see Chambers (2009). The WHO scale-up strategy includes identifying, documenting and assessing the nature of the innovation to be scaled up; increasing the capacity of the implementing agency; assessing the broader environment in which the project is to be scaled up; supporting the resource team which will support the scale up; embedding the project within the institutions of the target country; and documenting the scale up strategy.

Table 2
Does CLTS treatment improve sanitation and child health outcomes?

<i>Dependent Variables:</i>	Toilet Construction		Intolerance of Open Defecation		Diarrhea Knowledge	Roundworm(eggs/g)		Hemoglobin(g/l)	Weight z-score	Height z-score	Health Index
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Treatment	0.024 (0.014)*		0.409 (0.233)*		0.007 (0.058)	−72.81 (39.55)*		−0.18 (0.51)	0.03 (.03)	−0.01 (.03)	0.02 (.02)
RA Treatment* Poor		−0.027 (0.034)		0.644 (0.56)			−71.4 (110.5)				
RA Treatment* Non-poor		0.072 (0.028)***		1.035 (0.312)***			−176.4 (67.2)***				
LG Treatment* Poor		−0.028 (0.036)		0.069 (0.553)			35.9 (83.8)				
LG Treatment* Non-poor		0.014 (0.016)		−0.128 (0.316)			4.9 (48.4)				
Poor		0.009 (0.022)		−1.031 (0.359)***			−69.9 (65.9)				
Mean DV (Treat = 0)	0.125	0.125	32.9	32.9	4.85	156.7	159.2	111.0	−1.39	−1.65	0.001
Controls:	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Tests of Equality (p-values):											
RA* Non-Poor = RA* Poor		0.02		0.49			0.37				
LG* Non-Poor = LG* Poor		0.31		0.74			0.37				
RA* Non-Poor = LG* Non-Poor		0.06		0.004			0.01				
RA* Poor = LG* Poor		0.98		0.40			0.31				
RA* Poor = RA* Non-poor = LG* Poor = LG* Non-poor		0.09		0.04		0.68	0.09				
Observations	1858	1858	1858	1858	1858	1780	1742	1443	1886	1872	2043

Notes: We report results from OLS regressions (equations (1) and (2)). The dependent variables are: *Toilet Construction* which equals 1 if the household built a toilet since baseline and 0 otherwise; *Intolerance of Open Defecation* which is the sum of responses to 9 questions about attitudes toward open defecation (45 is the maximum score possible and is the highest level of intolerance while 9 is the minimum score possible and reflects total acceptance of open defecation); *Diarrhea Knowledge* which is a score out of 6 based on six questions about possible causes of diarrhea (a score of 6 indicates that the respondent got all of the questions correct); roundworm prevalence (eggs/g); hemoglobin (g/l); weight and height z-scores of children 0–5; and an index of roundworm, hemoglobin, weight z-scores, and height z-scores. RA (LG) treatment indicates villages are assigned to implementation by a resource agency (local government). Standard errors are clustered at the village level and are reported in parentheses. All specifications include sub-district fixed effects and household control variables (household size, the household head's age and educational attainment, household composition, log of per capita household income, eligibility for low income support and dwelling characteristics) and village control variables (the village population, village land area, the percentage of the village which is Muslim, whether there is a paved road to the nearest city, average years of education of household heads, whether a river flows through the village, and the percentage of households in the village who open defecated at baseline). Columns 6–11 also control for the sex of the child, and dummy variables for age in months of the child. ***indicates significance at 1% level, ** at 5% level, * at 10% level.

4. *Implementation agent effects.* The identity of the implementing agency may alter incentives.

In our context, many of these effects are not present. For example, as the program was implemented simultaneously by both RAs and LGs, any general equilibrium effects will be common to both treatment types. General equilibrium effects could include spillovers resulting from widespread increased demand for sanitation (e.g. latrine price changes and spillovers in information about the benefits of sanitation from treatment villages to control villages). In addition, the geographic and demographic context did not differ systematically and the project design is identical. Hence, it is a situation where there is a reasonable likelihood of successful scale up being achieved. The only potential difference between RA and LG implementation is in the *implementation agent* effect. That is the implementing actors and their associated administrative constraints differed.¹¹

Villages were not randomized into LG vs RA status, though discussions with WSP suggest there was nothing systematic about how these decisions were made. If the characteristics of villages differ with implementing agency then we could falsely ascribe differences in program effectiveness to LG versus RA triggering. Tests of whether villages that were assigned to be triggered by local governments are otherwise similar to the villages that were assigned to be triggered by resource agencies are presented in columns 4–6 of Table 1. Table 1 shows the villages are remarkably similar. There are no observable differences in the demographic and socio-economic composition of the villages. There are also no significant differences in access to sanitation or open defecation rates at baseline. This is important because if local governments are cherry-picking villages so as to work with communities that are most likely to become open defecation free then we would expect to see differences in baseline sanitation.

We also examine differences at the village level which might influence the population's interest in sanitation (panel C, Table 1). There are no differences in most of the village characteristics, including the accessibility of the villages (having a paved road to the nearest town and the distance to the city), levels of social capital, and whether a river runs through the village. Defecating in rivers is common practice in Indonesia and CLTS field workers report that motivating households to build toilets in villages that are on a river is more difficult (Mukherjee, 2011). There is also no difference in the percentage of households in the village that open defecate at baseline. The only difference we observe is that the RA-assigned villages have a significantly smaller population than LG-assigned villages. We control for village population in the specifications below. Population is not a significant determinant of the probability of building a toilet nor of the key health outcomes.

5.1.1. Poverty status

As discussed earlier, program impact may also vary with the poverty status of the household, as CLTS does not provide financial assistance in any form to poorer households, the ability of poorer households to participate in the program by constructing a toilet may be limited. Toilet construction requires a significant outlay of capital. In the endline

¹¹ The categorization here draws from and augments Grossman et al. (2015) who attribute the lower uptake in the scaled up version of the political engagement technology they study to a design effect—invitations to participate were given over the radio, rather than in person during a survey and an implementation agent effect—scale up involved implementation by parliament and promotion by politicians which may have altered incentives. The lack of replicability found in the teaching intervention in Bold et al. (2013) is slated to a combination of general equilibrium effects arising from political economy forces associated with union resistance to the hiring of a large number of contract teachers, and implementation agent effects. A further related study, Berge et al. (2012), examines implementation agency effects but in the context of a small-scale business training program in Tanzania. It finds that the training was much more effective when implemented by professional trainers rather than by a local NGO.

survey, cost is the most frequently reported obstacle to building a toilet, reported by 47% of households. Further, less than 5% of poor households report having sufficient savings to cover the cost of building a latrine (estimated by WSP to be USD 46).¹² Credit is rarely used as a financing mechanism, likely due to lack of availability and/or high interest rates. Only 2.5% of households who built a toilet report borrowing to do so.

We examine the heterogeneity of program impact by poverty status and implementer identity simultaneously. We generate an indicator to identify poor households with a household being deemed poor if they are in the bottom quartile of the distribution of non-land assets.¹³ We include this variable in the regressions interacted with treatment status and implementer identity.

5.1.2. Estimation strategy

Equation (2) is the estimating equation that allows for treatment effects to differ both by poverty status and the identity of the implementer. We note that we only test for heterogeneity among the dependent variables with a statistically significant average treatment effect (toilet construction, intolerance of open defecation, and roundworm). The outcome measures are regressed on a treatment dummy interacted with both implementing agency and poverty status. $T_{ij}^{RA*Poor}$ equals one if household i in village j is poor and village j was assigned to be triggered by an RA and zero otherwise; $T_{ij}^{RA*Nonpoor}$ equals one if household i in village j is not poor and village j was assigned to be triggered by an RA and zero otherwise. $T_{ij}^{LG*Poor}$ and $T_{ij}^{LG*Nonpoor}$ are defined analogously when village j is assigned to local government implementation. The estimating equation is:

$$Y_{ij} = \alpha + \beta_1 T_{ij}^{RA*Poor} + \beta_2 T_{ij}^{RA*Nonpoor} + \beta_3 T_{ij}^{LG*Poor} + \beta_4 T_{ij}^{LG*Nonpoor} + \gamma X_{ij} + \delta_K + \epsilon_{ij} \quad (2)$$

The coefficients of interest are β_1 , β_2 , β_3 , and β_4 . Comparisons of these coefficients reveals the differential impact of implementing agency for poor and less poor households. All other variables are as defined previously.

5.1.3. Results

Column 2 in Table 2 presents the results for toilet construction. Toilet construction increases significantly only among less poor households in treatment villages where the program was implemented by a resource agency. Non-poor households in these villages increased their toilet construction by 7.2 percentage points relative to non-poor households in control villages. The impact on toilet construction by poorer households in these treatment villages is not significant statistically (with a negative point estimate) and there are no significant impacts for households, whether poor or non-poor, in treatment communities where implementation was by local government. The difference in treatment impact between poor and non-poor households in RA treatment villages is statistically significant ($p=0.02$), as is the difference between the impact on non-poor households in RA and LG villages ($p=0.06$). We can also reject that all of the coefficients on the treatment variables are equal ($p=0.09$).

These results call into question CLTS's strategy of not providing subsidies for toilet construction since poorer households are no more likely to build toilets in treatment communities than in control communities, and the cost of construction is reported as a main obstacle to improving sanitation. In fact, recent empirical evidence suggests a crucial role

¹² See <http://millionssaved.cgdev.org/case-studies/indonesias-total-sanitation-and-sanitation-marketing-program>.

¹³ Note that most households in our sample are poor in the sense of being below the national poverty line. Here we are defining "poor" to capture the poorer households within our sample.

for subsidies (Dupas, 2014). Patil et al. (2014) and Pattanayak et al. (2009) evaluate India's Total Sanitation Campaign which uses a CLTS approach with subsidies. They find significant increases in access to improved sanitation but no robust health impacts. Hammer and Spears (2016) also study the Total Sanitation Campaign in the Indian state of Maharashtra and find the program has a large positive impact on children's heights. Guiteras, Levinsohn and Mobarak (2015) show that in Bangladesh, subsidies to the poor increase toilet ownership both among subsidized households and their unsubsidized neighbors, which suggests that investment decisions are interlinked across neighbors.

Column 4 presents results of the same specification but with the score for intolerance of open defecation as the dependent variable. Interestingly, the results for attitudinal change show a corresponding decrease in tolerance of open defecation among the non-poor in communities where a resource agency is the implementing agency, suggesting that the resource agencies are more effective at generating attitudinal change and that the program has difficulty affecting the attitudes of households who are least able to afford sanitation.

We also examine whether these differential improvements in sanitation infrastructure by triggerer identity and poverty status are apparent in the roundworm finding. Column 7, Table 2 shows that the reductions in roundworm are concentrated among less poor households in RA villages. In contrast, this coefficient for LG villages is not statistically significant (and close to zero). The difference between RA and LG villages is significant at the 1% level.¹⁴

5.1.4. How does RA implementation differ?

Table 3 compares facets of program implementation across treatment villages triggered by RA or LG. We investigate whether the way information was disseminated to the community (panel A); the extent of program engagement with village staff (panel B); the intensity of implementation and the use of rewards or competitions (panel C), and the extent of community participation (panel D) were different between RA and LG villages. We first present an overall measure for each panel and then include results for the individual variables that contribute to these measures in the subsequent columns. The summary measure for each overall indicator is defined in the table notes in Table 3.

There is no significant difference in the way the RAs and LGs disseminated information about the project (via TV, radio, print media, video, notices in shop windows or on village notice boards). The RAs are however more likely to engage with village staff, in particular, with the village office and with village health post volunteers. The intensity of implementation is greater in RA villages (driven by a greater number of facilitators visiting the communities, and facilitators making more visits). Most villages received only one visit from the team, some villages received two visits and a small number received three. RA facilitators made 0.42 more visits to villages than LG teams (significant at the 5% level). RA implementation also results in significantly greater community participation. Respondents in RA-triggered treatment villages are 13 percentage points more likely to have heard about the program and

12 percentage points more likely to have known about the triggering event.

In the field one hears a lot about the importance of the “quality” of the facilitator. In order to test whether the RA facilitators are “better” than the LG facilitators, we collected information from respondents on their perceptions of how charismatic/persuasive the facilitators were. We find no significant difference in the average reported persuasiveness of the facilitators (column 15). An examination of various program reports reveals that there was general satisfaction of WSP staff with the quality of the RA training of facilitators (Rosenzweig and Kopitopoulos, 2010).

5.2. Role of social capital

Given the community-led, participatory nature of CLTS it seems likely that initial levels of social capital in treatment villages would impact on program effectiveness. A higher level of social capital is thought to facilitate collective action by lowering the costs associated with such action (Casey et al., 2012); by reducing the problem of free riding; and facilitating the transmission of knowledge about the behavior of others and hence reducing the problems of opportunism (Collier, 1998). Communities with a greater degree of pre-existing community interaction are likely to be better prepared to cooperate and also have a greater store of community knowledge on which to draw when targeting the poor and prioritizing community needs. Conversely, communities with low stocks of social capital might struggle to work together and agree on community priorities.

High levels of social capital within a community may also result in households internalizing the social benefit of the provision of private goods, (Karlan, 2005). If a household builds a toilet and stops defecating in the village stream then other villagers benefit from this. If communities with higher levels of social capital internalize these social benefits more, they are likely to be more interested and willing to work together to improve the community's sanitation. Research on the relationship between initial levels of social capital and outcomes of participatory development programs is almost non-existent and where it does exist, data limitations mean that the direction of causality is not clearly identified.¹⁵ Reports from field workers suggest that the CLTS approach is particularly effective in settlements with a sense of community (Chambers, 2009).¹⁶

¹⁵ Isham and Kähkönen (1999) and Isham and Kähkönen (2002) use cross-sectional data from villages participating in community-based water interventions in Indonesia, India, and Sri Lanka, respectively, and find that higher levels of social capital are positively associated with greater household participation in the selection of the type of water infrastructure and construction monitoring, and that this can lead to greater health improvements. Pargal, Huq and Gilligan (1999) find that social capital is positively associated with voluntary solid waste management systems arising in Dhaka, Bangladesh. Evidence from case studies is mixed (Uphoff and Wijayarathna, 2000).

¹⁶ In addition to seeking to improve living standards in poor communities, participatory development projects are often explicitly viewed as a vehicle for building social capital. By empowering communities and providing a reason and process by which community members can work together for a common goal, participatory development provides a potential mechanism for increasing community member interactions, forging relationships and building trust (Avdeenko and Gilligan, 2015). In the long-run, gains in social capital might facilitate economic development and help to sustain program impacts (Dongier et al., 2003; Mansuri and Rao, 2004)). In results reported in a working paper now subsumed by this paper (Cameron et al., 2015) we explore this issue in the context of CLTS and find that the program did not build social capital. In fact, treatment reduced trust in already low social capital settings. For surveys of evidence on this issue see Wong (2012), Mansuri and Rao (2004), and Mansuri and Rao (2012).

¹⁴ The estimated poor performance of LGs relative to RAs in terms of toilet construction, attitudes towards open defecation and roundworm prevalence could reflect lesser adherence by the local governments to the treatment assignment. To investigate this possibility we estimated two stage least squares regressions where we instrument for whether a triggering (by either a RA or LG) was confirmed by community survey respondents as having taken place with whether the village was assigned to be a treatment village, and its interaction with whether the village was assigned to be triggered by a RA. This strategy allows for the differential effect on the probability of a treatment village actually being triggered depending on whether it was to be triggered by a resource agency or local government. The instruments are strongly predictive and the second stage results are consistent with the OLS results. Results are available upon request.

Table 3
Did RA implementation differ from LG implementation?

Panel A. Information Dissemination							
	Aggregate Indicator	TV	Radio	Print	Video	Shop Sign	Village Notice
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Treated by RA	0.05 (0.06)	−0.002 (0.05)	0.04 (0.03)	0.04 (0.05)	0.02 (0.01)	0.03 (0.03)	0.04 (0.04)
Mean DV	0.48	0.40	0.14	0.17	0.02	0.07	0.12
Observations	635	635	635	635	635	635	635
Panel B. Local Engagement							
	Aggregate Indicator	Health Officer	Midwife	Health Post Volunteers	Village Office		
	(8)	(9)	(10)	(11)	(12)		
Treated by RA	0.15 (0.06)**	0.09 (0.06)	0.06 (0.06)	0.11 (0.06)*	0.11 (0.04)***		
Mean DV	0.49	0.33	0.31	0.34	0.14		
Observations	635	635	635	635	635		
Panel C. Implementation Intensity							
	Aggregate Indicator	No. of Facilitators	Facilitator Charisma	No. of Visits	Rewards or Competition		
	(13)	(14)	(15)	(16)	(17)		
Treated by RA	0.45 (0.23)*	1.3 (0.77)*	0.16 (0.15)	0.42 (0.22)**	−0.04 (0.11)		
Mean DV	−0.07	3.19	2.8	0.91	0.10		
Observations	635	635	635	635	635		
Panel D. Community Participation							
	Aggregate Indicator	Heard about Program	Know about Triggering	Attended Triggering			
	(18)	(19)	(20)	(21)			
Treated by RA	0.34 (0.13)**	0.13 (0.06)**	0.12 (0.05)**	0.08 (0.05)			
Mean DV	0.90	0.66	0.16	0.10			
Observations	635	635	635	635			

Notes: The sample is restricted to observations in villages which were treated. We report the coefficient on the indicator that the village was treated by a resource agency (RA). *Information Dissemination Aggregate Indicator* equals 1 if any of TV, radio, print media, video, notices in shop windows or village notice boards were used to disseminate information about the program, 0 otherwise. The other dependent variables in Panel A equal 1 if the respondent heard about the sanitation program from the specified source, and 0 otherwise. In Panel B, *Local Engagement Aggregate Indicator* equals 1 if the program engaged with any of village health officers, midwives, health post volunteers or village officials, 0 otherwise. The dependent variables in columns 9–12 equal 1 if the program engaged with the specified officers, 0 otherwise. *Implementation Intensity Aggregate Indicator* is an unweighted standardized index of the variables in columns 14–17. *No. of Facilitators* is the village average of respondents reports of how many facilitators were at the triggering; *Facilitator Charisma* is the village average of respondent rankings from 1 to 4 of how charismatic/persuasive the facilitators was; *Number of visits* is the village average of respondent reports of how many visits the facilitators made to the village, and *Rewards or Competition* equals 1 if one or more respondents in the village reported that the program involved rewards for villages becoming open defecation free and/or competitions between villages with regard to decreasing open defecation. *Community Participation Aggregate Indicator* is the sum of the dependent variables in columns 19–21 which are indicators of whether the household had heard of CLTS, knew about the triggering and had attended the triggering. Standard errors are clustered at the village level. ***indicates significance at 1% level, ** at 5% level, * at 10% level.

5.2.1. Social capital empirical strategy and results

To examine the impact of social capital on the success or otherwise of the sanitation program we construct a village social capital index from data collected at baseline on participation in community groups and the extent of networks in the village. The index is constructed in the same way as we calculated the index of health outcomes above, following Kling et al. (2007). Table A2 in the appendix provides details of the household-level social capital variables collected at baseline that are used to construct the index. As we are interested in village level social capital, the index is calculated from village averages of each of these variables.

Table A2 shows that the social capital variables are balanced between treatment and control. There are no significant differences in any of the individual social capital variables, nor in the village social capital index. This is true across the entire sample and also within the sub-sample of households who do not have sanitation at baseline. We estimate the social capital regressions over households that had no access to sanitation at baseline so as to focus on households which can improve their sanitation through building a toilet in response to the program. In the previous sections when we were looking at both toilet construction and health outcomes, we examined estimates over the whole sample as health benefits may accrue to those who built a toilet as well as other households in the community. Table A3 additionally shows that access to sanitation, improved water sources and sanitation

behavior (handwashing) is balanced at baseline in the sub-sample for which we have social capital data.

We allow for program impact to differ with the level of baseline social capital (and implementing agency and poverty status) by adding three additional regressors to equation (2). The additional variables are the index of social capital at baseline; the index of baseline social capital interacted with a treatment dummy; and the index of baseline social capital interacted with RA treatment.

The estimating equation is:

$$\begin{aligned}
 Y_{ij} = & \alpha + \beta_1 T_j^{RA*Poor} + \beta_2 T_j^{RA*Nonpoor} + \beta_3 T_j^{LG*Poor} \\
 & + \beta_4 T_j^{LG*Nonpoor} + \beta_5 T_j * SC_{j[BL]} + \beta_6 T_j^{RA} * SC_{j[BL]} \\
 & + \beta_7 SC_{j[BL]} + \gamma X_{ij} + \delta_K + \epsilon_{ij}
 \end{aligned} \quad (3)$$

The new coefficients of interest are β_5 , β_6 , and β_7 . Everything else is as before.

The results from estimating equation (3) for toilet construction are reported in Table 4. We first show the average treatment effect for this smaller sub-sample. Column 1 shows that households in the social capital sample who did not have sanitation at baseline are 6 percentage points more likely to build a toilet than like households in control villages. Columns 2 and 3 include the interactions between treatment status, implementer identity, poverty status, and baseline village social capital (with and without controls). The level of social capital at

Table 4
Does social capital impact toilet construction?

	(1)	(2)	(3)	(4)	(5)
Treatment	0.06 (.02)***				
RA Treatment* Poor		0.07 (0.08)	0.10 (0.08)	−0.04 (0.15)	−0.003 (0.07)
RA Treatment* Non-poor		0.13 (0.04)***	0.11 (0.04)**	0.07 (0.05)	0.09 (0.05)*
LG Treatment* Poor		−0.01 (0.06)	−0.11 (0.06)*	0.01 (0.1)	0.007 (0.07)
LG Treatment* Non-poor		0.02 (0.03)	−0.02 (0.03)	−0.01 (0.04)	−0.008 (0.03)
Treatment* Village Social Capital BL		0.18 (0.08)**	0.14 (0.08)*		
RA Treatment* Village Social Capital BL		0.17 (0.15)	0.33 (0.16)**		
Treatment* Quintile 1 of Village Social Capital BL				−0.14 (0.06)**	−0.16 (0.04)***
Treatment* Quintile 2 of Village Social Capital BL				−0.02 (0.08)	
Treatment* Quintile 3 of Village Social Capital BL				0.04 (0.06)	
Treatment* Quintile 4 of Village Social Capital BL				−0.04 (0.06)	
Treatment* Quintile 5 of Village Social Capital BL				−0.06 (0.06)	
RA Treatment* Quintile 1 of Village Social Capital BL				−0.07 (0.1)	
RA Treatment* Quintile 2 of Village Social Capital BL				0.14 (0.09)	
RA Treatment* Quintile 3 of Village Social Capital BL				−0.01 (0.08)	
RA Treatment* Quintile 4 of Village Social Capital BL				0.37 (0.11)***	0.27 (0.09)***
RA Treatment* Quintile 5 of Village Social Capital BL				0.28 (0.09)***	0.18 (0.06)***
Village Social Capital BL		0.03 (0.14)	−0.11 (0.13)	−0.35 (0.12)***	−0.28 (0.11)***
Poor		−0.04 (0.03)	−0.02 (0.03)	−0.03 (0.03)	−0.03 (0.03)
Controls	No	No	Yes	Yes	Yes
Mean DV (Treatment = 0)	0.06	0.06	0.06	0.06	0.06
Test Q1 = Q2 = Q3 = Q4 = Q5				< 0.001	
Test RA* Q1 = RA* Q2 = RA* Q3 = RA* Q4 = RA* Q5				< 0.001	
Observations	596	596	596	596	596

Notes: These are OLS regressions on the sample of households that did not have access to sanitation facilities at baseline for the social capital sample from equations (1) and (3). All specifications include sub-district fixed effects. Standard errors clustered at the village level and are reported in parentheses. *** indicates significance at 1% level, ** at 5% level, * at 10% level. RA (LG) Treatment indicates villages assigned to implementation by a resource agency (local government). Village Social Capital BL is the baseline village social capital index constructed from the variables in Table A2. Column 5 drops the interactions with quintile of the baseline village social capital which are not statistically significant in column 4. Columns 3–5 include the usual set of controls – household control variables (household size, the household head's age and educational attainment, household composition, log of per capita household income, eligibility for low income support and dwelling characteristics) and village control variables (the village population, village land area, the percentage of the village which is Muslim, whether there is a paved road to the nearest city, average years of education of household heads, whether a river flows through the village, and the percentage of households in the village who open defecated at baseline).

baseline in treatment villages is strongly positively associated with the probability of toilet construction, particularly in RA treated villages. A one standard deviation increase in the baseline community participation index is associated with approximately an 11.3 percentage point (188%) increase in the probability that a household built a toilet (significant at the 5% level). This is a large increase and signifies substantial variation in program success dependent on the initial level of community participation. The program impacts are concentrated among the non-poor in RA villages with high levels of baseline social capital. A comparison of these results with and without controls establishes that the inclusion of controls does not substantially alter the estimated impacts.

Columns 4 and 5 in Table 4 allow for non-linearities in the impact of baseline village social capital. We interact treatment with indicators of quintiles of the distribution of the baseline social capital index. Toilet

construction is spurred by being in the top two quintiles of the social capital distribution in RA treatment villages. In villages with very low levels of community participation (lowest quintile) toilet construction is approximately 16 percentage points lower in treatment villages than in similar control villages. The linear model also predicts that fewer toilets are constructed in treatment communities than control communities when social capital is low. In the raw data, 11 percent of households constructed toilets in treatment communities in the lowest quintile of the social capital distribution compared to 20 percent in similar control communities.

5.2.2. Mechanisms for the social capital result

We test three hypotheses to better understand what might be driving the social capital results. First, a more active community may mean a better informed community as members know each other better and

Table 5
Role of baseline social capital in treatment communities.

Dependent Variable:	Information Index EL(1)	Sharing/Attendance Index EL(2)	Sanctions Index EL(3)
Village Social Capital Index BL	0.13 (0.13)	0.09 (0.15)	0.51 (0.18)***
Control variables	Yes	Yes	Yes
Sub-district fixed effects:	Yes	Yes	Yes
Mean DV (treatment = 0)	0.01	0.00	0.00
Observations	721	721	721

Notes: These are OLS regressions for the entire sample of treatment households for which we have social capital data. Village Social Capital BL is the baseline village social capital index constructed from the variables in Table A2. All regressions include household control variables (household size, the household head's age and educational attainment, household composition, log of per capita household income, eligibility for low income support and dwelling characteristics) and village control variables (the village population, village land area, the percentage of the village which is Muslim, whether there is a paved road to the nearest city, average years of education of household heads, whether a river flows through the village, and the percentage of households in the village who open defecated at baseline). Standard errors clustered at the village level and are reported in parentheses. ***indicates significance at 1% level, ** at 5% level, * at 10% level.

exchange information when they meet. To examine this hypothesis we construct an index of information shared within the community at end-line from variables reflecting whether the household reported that they knew about the triggering event; whether they learned about sanitation construction from other community members; and whether knowledge about the causes of diarrhea increased between baseline and end-line.¹⁷ Whether villages with higher levels of baseline participation have greater information flows at endline is tested by regressing this index on baseline village social capital. Baseline social capital is not associated with significantly greater information flows (see Table 5, column 1).

Second, in more active communities, people may be more willing to be actively involved and share resources as a result of knowing each other better. In the CLTS context this may result in being more likely to attend the triggering event and more shared and public toilets being built. To test this hypothesis we construct an index from these two variables and examine its relationship with the baseline social capital index. Baseline social capital is not a significant determinant of active involvement and sharing of resources (see Table 5, column 2).¹⁸

The final mechanism we examine is whether social sanctions play a greater role in encouraging households to build toilets in communities with more social capital. If household members know their community better, they may be more concerned about what other community members think of them. We construct an index of sanctions from reports on a scale of 1 (strongly agree) to 5 (strongly disagree) on whether building a toilet reduces the likelihood of being a target of gossip; whether those who defecate in the open will not be accepted by the community; and whether the community imposes social sanctions on those who defecate in the open. Table 5 (column 3) shows that villages with higher levels of social capital at baseline are more likely to impose sanctions, consistent with these communities being more able to regulate behavior by the use of social opprobrium.

6. Conclusion and policy implications

CLTS modestly increased the rate of toilet construction and reduced community tolerance of open defecation. We find an associated decrease in roundworm infestations but no improvements in children's hemoglobin levels, weight or height. An index of child health also shows no significant overall improvement. Although the rate of toilet construction increased about four percentage points among less poor

households, the poorest households did not build toilets. This highlights potentially important roles for the provision of finance to poor households and/or subsidies for the poor in conjunction with CLTS in producing open defecation free communities (and the possible concomitant health benefits).

The examination of the scale up process shows that CLTS had relatively large positive impacts in villages where the program was implemented by RAs. In contrast, with the identical program design, the same demographic composition of participating households, and common general equilibrium effects, implementation by local governments failed to produce any discernible positive impacts. Understanding what makes for successful scale up is of prime importance to the development sector. Currently there are very few studies that explicitly examine the scale up process through the lens of a rigorous quantitative evaluation and the studies that exist find either a lack of replicability at scale or that successful scale up is not straightforward and involves considerable learning from failure. Integration of quantitative evaluation, qualitative research, and high quality monitoring data is likely to improve researcher and program implementers' ability to understand the causes of success and failure, so as to increase the likelihood of successful scale up in the future.¹⁹

Finally, our results show that CLTS increased toilet construction in villages with sufficiently high pre-existing social capital in the form of community participation. In villages with low initial levels of social capital, however, the program was counterproductive—resulting in fewer toilets being built. We present evidence consistent with high social capital communities being better able to use social pressure to get community members to conform with program objectives. Our findings are thus cautionary with respect to using participatory development approaches in low social capital environments and at the very least suggest a need for greater investment in community-support for participatory development programs in areas with demonstrably low social capital.

¹⁷ The method of Kling et al. (2007) is used to construct the indices in Table 5.

¹⁸ We also directly examine the construction of public and shared toilets. More active communities do not build more public and shared toilets.

¹⁹ For discussion of data collection efforts to identify implementation and scale up difficulties see Bamberger et al. (2010); Pritchett et al. (2013); Andrews et al. (2012); Woolcock (2013); Kruk et al. (2016); Brown et al. (2014). Banerjee et al. (2016) used process data to identify the keys to scale up success.

Table A1

Does CLTS improve sanitation and child health outcomes? (No Controls).

Dependent Variables:	Toilet Construction		Intolerance of OD		Diarrhea Knowledge	Roundworm (eggs/g)		Hemoglobin (g/l)	Weight z-score	Height z-score	Health Index
	(1)	(2)	(3)	(4)		(6)	(7)				
Treatment	0.03 (0.01)**		0.28 (0.28)		−0.01 (0.05)	−58.9 (35.5)*		0.07 (0.45)	0.02 (0.03)	−0.03 (0.03)	0.03 (0.02)
RA Treatment* Poor		−0.03 (0.03)		0.78 (0.64)			−39.7 (105.9)				
RA Treatment* Non-poor		0.08 (0.03)***		1.21 (0.37)***			−151.3 (64.2)**				
LG Treatment* Poor		−0.01 (0.03)		−0.39 (0.59)			35.0 (77.4)				
LG Treatment* Non-poor		0.03 (0.02)**		−0.47 (0.4)			7.6 (43.1)				
Poor		0.008 (0.02)		−1.93 (0.36)***			−80.7 (69.1)				
Mean DV (Treat = 0)	0.125	0.125	32.9	32.9	4.85	156.7	159.2	111.0	−1.39	−1.65	0.001
Controls:	No	No	No	No	No	No	No	No	No	No	No
Tests of Equality (p-values):											
RA* Non-Poor = RA* Poor		0.01		0.49			0.33				
LG* Non-Poor = LG* Poor		0.29		0.90			0.77				
RA* Non-Poor = LG* Non-Poor		0.15		0.002			0.03				
RA* Poor = LG* Poor		0.62		0.14			0.45				
RA* Poor = RA* Non-poor = LG* Poor = LG* Non-poor		0.07		0.02			0.16				
N	1858	1858	1858	1858	1858	1780	1742	1443	1886	1872	2043

Notes: We report results from OLS regressions (equations (1) and (2)). The dependent variables are: *Toilet Construction* which equals 1 if the household built a toilet since baseline and 0 otherwise; *Intolerance of Open Defecation* which is the sum of responses to 9 questions about attitudes toward open defecation (45 is the maximum score possible and is the highest level of intolerance while 9 is the minimum score possible and reflects total acceptance of open defecation); *Knowledge of Causes of Diarrhea* which is a score out of 6 based on six questions about possible causes of diarrhea (a score of 6 indicates that the respondent got all of the questions correct); roundworm prevalence (eggs/g); hemoglobin (g/l); weight and height z-scores of children 0–5; and an index of the roundworm, hemoglobin, weight and height z-scores. RA (LG) treatment indicates villages are assigned to implementation by a resource agency (local government). Standard errors are clustered at the village level and are reported in parentheses. ***indicates significance at 1% level, ** at 5% level, * at 10% level.

Table A2
Social Capital Measures.

Variables	All Households			No Sanitation at Baseline		
	Mean (Treatment)	Mean (Control)	p-value Difference	Mean (Treatment)	Mean (Control)	p-value Difference
Did a household member participate in a religious group in the last 12 months?	0.77	0.76	0.74	0.71	0.76	0.28
Did a household member participate in a women's group in the last 12 months?	0.57	0.54	0.63	0.57	0.54	0.73
Did a household member participate in a rotating savings (arisan) group in the last 12 months?	0.72	0.65	0.15	0.68	0.64	0.47
How many close friends do you have in the community?	5.49	5.66	0.68	5.58	5.45	0.83
If you suddenly needed to borrow money (without interest) to meet household expenses is there someone in the community (other than family) who would be prepared to help you?	2.50	2.54	0.58	2.51	2.51	0.95
(Yes = 3/Maybe = 2/No = 1)						
Village Social Capital Index	0.01	−0.02	0.47	−0.03	−0.01	0.71
Observations	721	728		289	307	

Notes: This table shows the means for each of the variables which are used to generate the village social capital index, for all households as well as for the sub-sample of households that did not have private sanitation facilities at baseline. It also presents the means of the index. The p-values in columns 3 and 6 are generated from tests of statistical difference between treatment and control communities. *** indicates difference is significant at 1% level, ** at 5% level, * at 10% level.

Table A3
Balance Table – Social Capital Sample.

Variables	All Households			No Sanitation at Baseline		
	Mean (Treatment)	Mean (Control)	p-value Difference	Mean (Treatment)	Mean (Control)	p-value Difference
<i>Baseline Sanitation Variables:</i>						
Do not have own sanitation facility	0.52	0.51	0.89	1.00	1.00	.
Open defecate	0.40	0.42	0.74	1.00	1.00	.
Access to unimproved sanitation	0.11	0.14	0.37	0.00	0.00	.
Access to improved sanitation	0.49	0.43	0.27	0.00	0.00	.
Wash hands after defecation	0.99	0.99	0.52	0.97	0.99	0.31
Access to piped water	0.04	0.07	0.37	0.03	0.04	0.69
Intolerance of Open Defecation	33.3	32.6	0.27	30.3	30.0	0.64
Believes open defecation causes diarrhea	0.68	0.65	0.36	0.60	0.52	0.10*
Knowledge of causes of diarrhea	4.64	4.61	0.82	4.29	4.24	0.78
<i>Baseline Household Characteristics:</i>						
Household head's age	40.23	40.12	0.89	40.11	38.45	0.13
Household head male	0.95	0.97	0.14	0.95	0.99	0.05*
<i>Household head's educational attainment:</i>						
Elementary	0.51	0.51	0.85	0.63	0.63	0.87
Lower secondary	0.20	0.20	0.75	0.17	0.17	0.93
Upper secondary	0.17	0.20	0.47	0.11	0.10	0.91
Tertiary	0.05	0.04	0.43	0.01	0.01	0.43
Household size	4.94	4.82	0.26	4.91	4.64	0.07*
<i>Number of children in the household:</i>						
Aged 0–5 years	1.15	1.13	0.33	1.13	1.07	0.01***
Aged 6–10 years	0.35	0.30	0.13	0.37	0.28	0.05*
Aged 11–17 years	0.37	0.38	0.78	0.41	0.41	0.93
Per capita household income (mill Rp/year)	2.78	2.96	0.51	1.77	1.85	0.61
House has a dirt floor	0.22	0.24	0.60	0.32	0.32	0.95
House has a tiled floor	0.05	0.06	0.49	0.03	0.04	0.82
House has walls of brick or wood	0.89	0.88	0.83	0.80	0.82	0.61
Household uses wood as a cooking fuel	0.57	0.55	0.59	0.69	0.63	0.30
Cash transfer program (BLT) recipient	0.24	0.26	0.61	0.39	0.37	0.63
<i>Baseline Village Characteristics:</i>						
Village population	925.61	1052.51	0.21	835.33	894.69	0.46
Paved road to the nearest city	0.92	0.95	0.46	0.92	0.96	0.52
% of the village population that are muslim	97.72	97.26	0.68	98.61	98.41	0.77
River runs through village	0.83	0.86	0.62	0.92	0.93	0.86
% of households in village that open defecate	15.18	14.66	0.86	24.14	21.27	0.45
Average years of education of household heads	10.13	10.45	0.26	9.67	10.40	0.06*
Village land area	63,297	33,543.05	0.27	109,804	22,601.86	0.03**
Observations	721	728		289	307	

Notes: These are summary statistics (means) using the baseline data from the social capital sample. *Intolerance of Open Defecation* is the sum of responses to 9 questions about attitudes toward open defecation (45 is the maximum score possible and is the highest level of intolerance while 9 is the minimum score possible and reflects total acceptance of open defecation); *Knowledge of causes of diarrhea* is a score out of 6 based on six questions about possible causes of diarrhea (a score of 6 indicates that the respondent got all of the questions correct). The p-values are generated from tests of statistical difference between treatment and control communities. *** indicates difference is significant at 1% level, ** at 5% level, * at 10% level.